Soft Skills, Hard Times: Why New College Graduates Are Falling Behind *

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Abstract

The labor market outcomes for young college graduates have been worsening since the Great Recession. We document that this secular trend coincides with a rise in demand for soft skills by employers. We propose that this shift in skill demand disproportionately affects young workers, who have had less opportunity to accumulate these skills through work experience. To quantify this mechanism, we build a general equilibrium search model with heterogeneous agents and on-the-job skill accumulation, calibrated to the US labor market. In this model, a rise in demand for soft skills explains about one-fifth of the increase in the unemployment rate of young workers relative to older workers.

Keywords: soft skills, youth unemployment, experience premium
JEL classification: J23, J24, J64

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Young college graduates have historically enjoyed substantially lower unemployment than the US workforce as a whole. However, since 2011, that advantage has gradually disappeared. This paper proposes that part of this deterioration reflects a change in the skills demanded by employers. Lindqvist and Vestman (2011), Deming (2017) and Edin et al. (2022) document a rising importance of soft skills in the labor market. We argue that many of these skills are accumulated through work experience. While college graduates may enter the labor market with adequate technical training, they have limited opportunities to develop these experience-dependent skills. As employers demand more soft skills, young college graduates find it increasingly difficult to compete with older, more experienced workers.

Using the Current Population Survey (CPS) and data on the near-universe of online job postings in the US obtained from Lightcast, we document three empirical facts that motivate our hypothesis. First, we show that historically, young college graduates were less likely to be unemployed than the average US worker.[1] However, we find that since 2011, the unemployment rate of young college graduates has been rising, and since 2017, they have been more likely to be unemployed than the average US worker. We further show that this deterioration is concentrated in their relative job-finding rates. In contrast, their relative separation rates have changed little. This suggests that young graduates' worsening labor market outcomes reflect a growing disadvantage in finding employment rather than an increase in separations.

Second, we show that coinciding with the rising unemployment rate of young college graduates, there has been an increase in the demand for soft skills by employers in the US labor market. Following O*NET, we define soft skills as interpersonal and thinking skills needed to interact with others and perform effectively in the workplace. Using this definition, we distinguish between soft and technical skills in our data through a combination of large language models and manual review.[2]

For example, a job posting for an engineer at an information technology (IT) firm requires an applicant to be familiar with multiple programming languages and software (which we categorize as technical skills), but it also asks the applicant to have strong communication and teamwork skills (which we identify as soft skills). Working with this classification, we find that between 2011 and 2024, technical skill demand increased by 6%. However, over the same period, soft skill demand increased more than twice as fast, by 16%. Moreover, this increase in soft skill demand is not limited to a few occupations but is present in most occupations in our job postings dataset. Using the full text of job postings, we show that not only has the share of jobs requiring substantial use of soft skills increased, employers are increasingly demanding both types of skills together, suggesting that technical expertise alone may no longer be sufficient to meet employers' requirements.

Third, we provide evidence that young college graduates experience worse outcomes in occupations that require more soft skills. Specifically, one additional soft skill per job posting in an occupation is associated with about 3% lower wages for young college graduates relative to older college graduates. In contrast, this wage gap becomes narrower in occupations that

[1] We define young workers as those aged 25 or younger. Older workers are aged 26 to 55.

[2] We measure soft (technical) skill demand as the average number of soft (technical) skills required per job posting. We also use a large language model to measure the intensity of soft- and technical-skill requirements in job advertisements.

require more technical skills (a 1% narrower wage gap per additional technical skill required).

These wage gap patterns suggest that although young college graduates have the requisite level of technical skills, they lack the soft skills required by employers. This can happen if soft skills are primarily accumulated on the job, which would imply that younger workers have had less opportunity to acquire them than older workers. Together, our three empirical facts suggest that an increase in the demand for soft skills may be an important reason why young college graduates are struggling in the labor market.

Why has the demand for soft skills increased? Autor et al. (2003) and Atalay et al. (2020) document that the task content of work has changed: as automation replaces workers in routine tasks, the tasks left for workers become more specialized, requiring workers to coordinate with one another to produce output (Garicano, 2000; Deming, 2017). Through a lab experiment, Weidmann and Deming (2021) show that workers with strong social skills consistently cause their teams to exceed their predicted performance, indicating the value of soft skills in the workplace. Therefore, as automation makes work more specialized, the importance of communication and teamwork grows, and employers demand more soft skills.

Motivated by our empirical facts, we use a general equilibrium search-and-matching model to quantify how much of the rise in the unemployment rate of young college graduates can be explained by the rise in demand for soft skills. Our model contains heterogeneous workers and firms, with on-the-job search and aggregate productivity shocks. New entrants start unemployed and with relatively low skills. Firms post vacancies, and both employed and unemployed workers receive job offers at random. Match creation is efficient, and wage determination happens through the sequential auctions framework developed by Postel-Vinay and Robin (2002). Most importantly, workers can accumulate skills on the job, resulting in a positive correlation between skill levels and work experience. Thus, young workers enter the labor market with relatively low skills, while older workers have had more opportunities to accumulate them through employment.

In equilibrium, the match surplus determines which matches form and how workers move between firms. Firms post vacancies according to the expected surplus of a match, and their vacancy creation decision depends both on workers' skills and on how many workers of each type are searching. There is complementarity between firm types and worker skills, which implies that low-skilled workers are only able to match with low-type firms, whereas high-skilled workers can climb the job ladder to match with high-type firms. Consequently, an increase in the returns to worker skills can change not only the total number of vacancies created, but also the types of firms creating these vacancies. This distinction is important because an increase in aggregate vacancy creation need not improve job-finding opportunities equally for all workers.

We estimate this model by matching key aggregate moments from the US labor market between 1951 and 2011. Our model matches the average rates of unemployment and labor market transitions well during this time period. Firm heterogeneity and the vacancy cost function parameters are disciplined by the cross-sectional distribution of firms' value added and time-series data on vacancies, respectively. Production function parameters determining the complementarity between firm types and worker skills are disciplined by correlations between labor market outcomes over time. Average wage growth over 30 years of work experience gov-

erns the speed of skill accumulation in the model.

In our counterfactual exercise, we model an increase in skill demand by incorporating a labor productivity shock equivalent to the rise in soft skill demand that we observe in the data over the 2011-2024 period. Since worker skills and firm types are complements in our estimated model, high-type firms find it more profitable to create vacancies, and the vacancy distribution shifts rightwards. Moreover, due to sorting in the labor market, low-skilled (young) workers can only form viable matches with low-type firms. Consequently, these young, low-skilled workers face a lower share of viable vacancies after the labor productivity shock hits, and their unemployment rate rises. We show that this calibrated increase in skill demand generates approximately one-fifth of the observed increase in the unemployment rate of young college graduates relative to older college graduates.

Past research documents the declining returns to cognitive skills and the rising importance of noncognitive (soft) skills. Castex and Kogan Dechter (2014) document that cognitive skills had a 30-50% larger effect on wages in the 1980s than in the 2000s. Beaudry et al. (2016) and Brynjolfsson and McAfee (2014) attribute this to the rise in the adoption of IT, as computers can substitute for human workers by easily doing tasks requiring cognitive skills. During the same time period, some papers show that noncognitive skills have become an important determinant of labor market outcomes. Using Swedish military enlistment data, Lindqvist and Vestman (2011) observe that low levels of noncognitive skills are a strong predictor of failure in the labor market (i.e., unemployment), while high levels of cognitive skills are necessary for success (i.e., high wages). Moreover, Deming (2017) in the case of the US, and Edin et al. (2022) in the context of Sweden show that the returns to noncognitive skills have increased over time. Our paper contributes to this literature by examining how rising soft-skill demand affects workers with limited opportunities to accumulate these skills through employment. While the existing literature documents the increasing value of soft skills, we study how this change can affect the employment opportunities of young workers relative to more experienced workers.

Another set of papers discusses wage stagnation and the reduced demand for junior talent. Guvenen et al. (2022) document that median lifetime earnings in the US stagnated between 1980 and 2013. They show that this occurs primarily due to the stagnation of wages at the time of labor market entry. The New York Federal Reserve keeps track of the labor market outcomes of young workers, and it shows that wages have remained stagnant in recent years as well. Bianchi and Paradisi (2024) and Citino et al. (2026) document the difficulties faced by junior talent, which have intensified in recent times due to remote work (Schubert, 2026; Emanuel et al., 2026). Our analysis suggests that rising demand for skills accumulated through employment may be another reason why young workers have fallen behind. Unlike explanations based on slower labor demand or constraints on career advancement, we emphasize how changing skill requirements affect firms' vacancy creation and young workers' job-finding opportunities.

Brynjolfsson et al. (2025) and Hosseini Maasoum and Lichtinger (2025) attribute the worsening outcomes of young college graduates to AI, while Emanuel et al. (2026) propose remote work as an explanation. AI can shift employer demand towards older workers, since AI may not have the context-specific knowledge that comes with experience, although it can easily do intensive cognitive tasks that are usually assigned to young workers. Remote work can

harm the productivity of young workers, as Emanuel et al. (2026) show that proximity to experienced workers is important for young, inexperienced workers to learn how their firm operates. Even though both AI and remote work are important determinants of young workers' outcomes in recent years, these reasons by themselves cannot explain why this worsening trend exists even in pre-COVID times. Our paper proposes another mechanism at play: the rising demand for soft skills that are primarily accumulated on the job. This skill demand mechanism should be considered in conjunction with the AI and remote work mechanisms to explain the current difficulties faced by young workers.

Our paper is also related to Wright and Wurdinger (2026). They document an almost 50% decline in the employment rate of high school students since the 1990s, attributing it to crowding out by adult workers whose own labor market opportunities have worsened. While both papers study the difficulties faced by young workers in accumulating work experience, we focus on young college graduates and a different mechanism. Rather than crowding out by adult workers, we examine how rising demand for accumulated skills changes firms' vacancy creation decisions, making some jobs less accessible to young workers.

Finally, our paper also builds on the literature studying worker-firm sorting and human capital accumulation in frictional labor markets. Lise and Robin (2017) develop a search-and-matching model with heterogeneous workers and firms, endogenous vacancy creation, and aggregate productivity shocks. Lise and Postel-Vinay (2020) study multidimensional skill accumulation and sorting across jobs with different skill requirements. Building on these frameworks, we examine how an increase in the returns to skills accumulated through employment changes equilibrium vacancy creation. We show that although higher returns to accumulated skills can increase overall vacancy creation, they can also reduce the share of viable vacancies available to inexperienced workers, worsening their relative unemployment outcomes.

The rest of the paper is organized as follows. First, we describe our data sources and the classification of soft and technical skills in Section I. We then document our empirical results in Section II. In Section III, we describe our model, and discuss its parametrization in Section IV. Section V contains our counterfactual exercise. We conclude in Section VI.

I Data

To understand the labor market outcomes of young college graduates, we use data from the Current Population Survey (CPS) to measure worker outcomes, and online job postings data from Lightcast to measure employer demand for skills. We also use occupational data from the Occupational Information Network (O*NET) to assess the skill measures. Appendix A1 describes the data construction and reports additional descriptive tables and figures.

I.A Workers and Labor-Market Outcomes

The basic monthly CPS allows us to measure unemployment and follow workers' transitions into and out of employment. For earnings, we use the merged outgoing rotation group (MORG) data. The outgoing rotation groups consist of households in their fourth or eighth CPS interview, when the survey asks additional questions about usual hours and earnings. Our de-

scriptive sample compares college graduates aged 22–25 with those aged 26–55, where college graduates hold a bachelor's degree or higher.[3] The civilian sample contains 9.2 million person-months with observed education and includes part-time workers, the self-employed, and people outside the labor force. We use final CPS survey weights and calculate unemployment as a share of the labor force. Over 2011–2024, unemployment averages 6% among young college graduates and 3% among older graduates. Table A1 reports the full descriptive statistics.

I.B Job Postings and Skill Requirements

Lightcast, formerly Burning Glass Technologies, collects online job postings and extracts information on the employer, occupation, and skills requested. These data allow us to follow changes in skill requirements within occupations. Our sample contains about 307 million postings from 2011–2024 across 769 occupation codes. We exclude internships, military occupations, and postings missing the information needed for the analysis.[4]

We distinguish soft from technical skills using O*NET's definition of soft skills:

> *"[I]nterpersonal and thinking skills needed to interact successfully with people and to perform efficiently and effectively in the workplace."*
>
> — National Center for O*NET Development (2026)

This category includes communication, management, negotiation, planning, and problem solving, as well as self-motivation and attention to detail. Technical skills concern the knowledge, tools, and procedures used to carry out a job, such as programming, accounting, clinical methods, and operating equipment. We classify the Lightcast skill labels using the taxonomy rules in Appendix A1.2; all labels outside the soft-skill set enter the technical category, including certifications and general workplace tools. Our results are also robust to using the classification in Deming (2017).

Our measure of skill demand is the average number of distinct skills required per posting in an occupation and month, calculated separately for soft and technical skills. The measure therefore captures the breadth of requirements, giving equal weight to each skill.[5] Table 1 illustrates common skill labels and their frequencies.[6] Communication appears in 34% of postings and management in 26%, while writing appears in 11% and Microsoft Excel in 8%.

Figure 1 shows the variation in these requirements across occupations typically requiring a bachelor's degree. Public relations specialists list more soft skills than accountants and auditors, while software developers list more technical skills. Appendix Figure A2 shows the corresponding map for all occupations.

[3] Age identifies young degree holders, rather than graduation dates or actual work experience. The age ranges and weights used in the wage regressions and transition analysis are reported in Appendix A1.1 and the notes to the corresponding exhibits.

[4] We require the posting date, occupation, industry, job title, employer, and skill list. Appendix A1.2 describes the posting restrictions.

[5] O*NET separately rates the importance and level of skills required in each occupation. Appendix A3 compares these ratings with our skill counts.

[6] The table uses an earlier classification of individual labels based on LLMs and manual review, which differs from the taxonomy rules used in the skill-count figures.

Table 1: Most Common Soft and Technical Skills in Lightcast Job Postings, 2011–2024

Panel A: Soft Skills

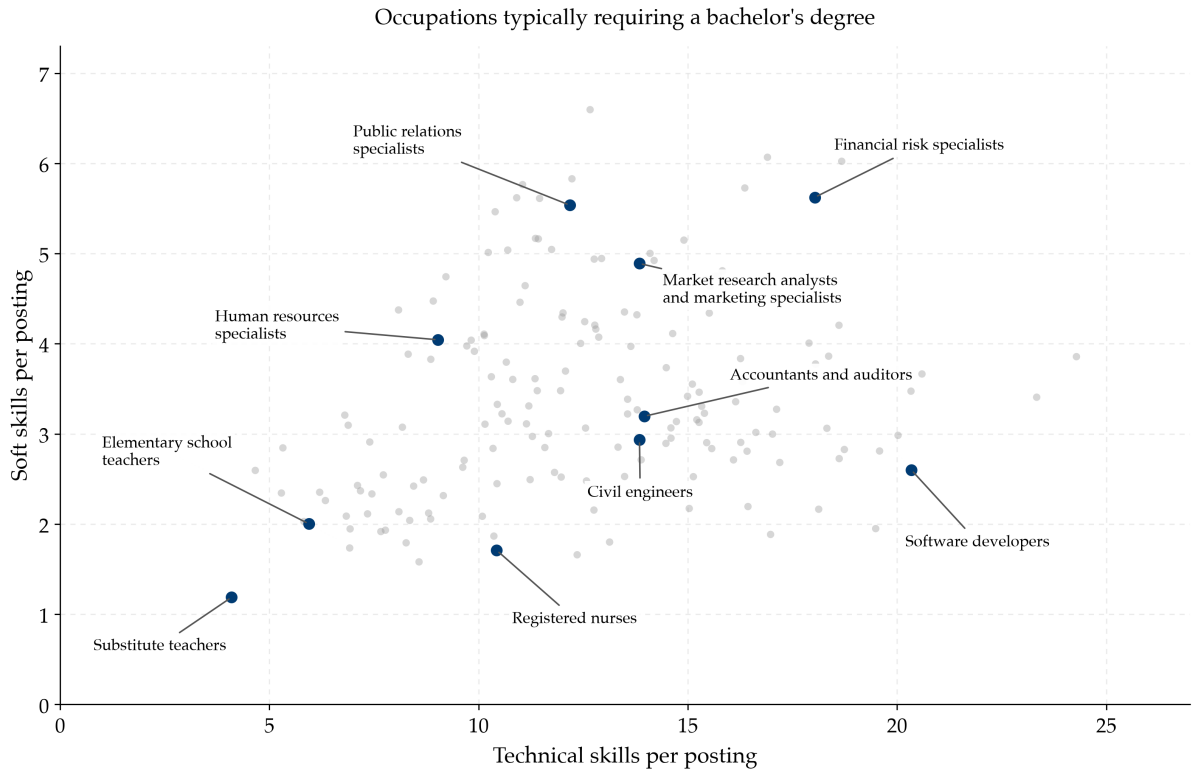
#	Skill	Share (%)	#	Skill	Share (%)
1	Communication	33.76	16	Time Management	5.72
2	Customer Service	28.97	17	Presentations	5.64
3	Management	25.59	18	Troubleshooting (Problem Solving)	5.55
4	Sales	19.70	19	Professionalism	5.29
5	Operations	16.93	20	Selling Techniques	5.12
6	Leadership	15.63	21	Teamwork	4.99
7	Problem Solving	11.51	22	Prioritization	4.46
8	Detail Oriented	11.48	23	Coaching	4.44
9	Planning	10.16	24	Decision Making	3.98
10	Interpersonal Communications	8.18	25	Cleanliness	3.95
11	Coordinating	6.96	26	Innovation	3.87
12	Multitasking	6.83	27	Accountability	3.68
13	Verbal Communication Skills	6.78	28	Influencing Skills	3.23
14	Organizational Skills	6.71	29	Analytical Skills	3.12
15	Self-Motivation	6.33	30	Mentorship	2.99

Panel B: Technical Skills

#	Skill	Share (%)	#	Skill	Share (%)
1	Writing	10.87	16	Microsoft PowerPoint	4.32
2	Microsoft Excel	8.48	17	Mathematics	4.25
3	Valid Driver's License	8.33	18	Warehousing	4.05
4	Registered Nurse (RN)	8.15	19	Basic Life Support (BLS) Certification	4.04
5	Project Management	7.66	20	Microsoft Outlook	4.03
6	Merchandising	7.43	21	Computer Science	4.02
7	Microsoft Office	7.41	22	Restaurant Operation	3.77
8	Research	7.01	23	Workflow Management	3.04
9	Marketing	6.76	24	SQL (Programming Language)	2.98
10	Computer Literacy	6.62	25	Microsoft Word	2.89
11	Nursing	6.61	26	Product Knowledge	2.82
12	Accounting	6.08	27	Process Improvement	2.80
13	Auditing	5.85	28	Billing	2.80
14	Scheduling	5.77	29	Invoicing	2.74
15	Finance	4.98	30	Cash Register	2.71

Notes: This table lists the 30 most common skills in each category in Lightcast postings from 2011–2024. The categories come from an earlier classification of individual skill labels using LLMs and manual review. They differ from the taxonomy rules used in the skill-count figures, described in Appendix A1.2. Each share is the percentage of postings requesting that skill. The separate LLM intensity measure assigns high or low ratings to the requirements described in a full advertisement.

Figure 1: Soft and Technical Requirements across College-Entry Occupations



Notes: Each point represents one of 169 Lightcast occupations for which the Bureau of Labor Statistics lists a bachelor's degree as the typical entry requirement in its 2025 education and training assignments. Occupations requiring postgraduate degrees are excluded. The horizontal and vertical axes show the average numbers of technical and soft skills per posting, respectively, pooling 118,730,974 postings from 2011–2024. The restriction applies to occupations; the averages use all postings within those occupations. Skill mentions and postings are summed within occupation before calculating ratios, giving each posting equal weight. Soft skills follow the taxonomy rules in Appendix A1.2; technical skills comprise all remaining labels, including certifications. Blue points mark ten illustrative occupations with no prior related work experience typically required according to BLS.

For the wage analysis, we assign the average soft and technical requirements in each occupation and month to CPS respondents in that occupation and month, retaining only workers for whom these averages are available. We first map the occupation codes in the two sources to common groups, summing skill mentions and postings within each group before calculating the averages. This allows us to examine how the wage gap between young and older graduates varies with the skills employers demand. The descriptive wage sample in the CPS includes wage and salary workers with at least 35 hours at their main job and positive earnings and earnings weights. We calculate hourly earnings from weekly earnings and hours and express them in 2024 dollars using the personal consumption expenditures price index.[7] Mean hourly earnings are about $28 for young college graduates and $43 for older graduates.

The count measure gives the same weight to a generic request for communication skills as to a requirement to negotiate contracts. We therefore complement it with a measure of skill intensity based on the full text of the advertisement. We use a large language model (LLM) to classify about 100,000 Lightcast postings from 2012–2025, assigning separate high or low ratings to soft and technical requirements. A high rating requires that the skill type be important for the main duties, repeatedly emphasized, or explicitly required at an advanced level. Generic requests for good communication or teamwork receive a low rating; leading a team or negotiating with clients can support a high soft-skill rating. A posting can be rated high in both dimensions. The model receives O*NET occupational ratings as context but must base its assessment on the advertisement's duties and qualifications. Appendix A2 describes the sample, prompt, and construction of the series.

We use these data to examine whether changing skill requirements can help explain the worsening labor market outcomes of young college graduates. The next section begins with their unemployment and job-finding rates, then documents changes in employers' demand for soft and technical skills. We also examine how these requirements relate to the wage gap between young and older graduates, and whether the experience premium has increased across cohorts.

II Empirical Analysis

College prepares graduates for many of the technical requirements of work, but they enter the labor market with little experience. Our hypothesis is that employers increasingly demand soft skills developed on the job, leaving recent graduates at a disadvantage despite their technical training.

Unemployment and Job Finding. The unemployment rate of young college graduates has seen a reversal and has been above the overall unemployment rate since 2017. Figure 2 compares the unemployment rate of graduates aged 25 or younger with the overall rate. Young graduates generally had lower unemployment before 2011. However, this advantage has subsequently narrowed, and their unemployment rate exceeded the overall rate by 2017, before

[7] We use reported hourly pay when the ratio of weekly earnings to hours is unavailable. The descriptive statistics retain allocated and top-coded earnings and use ORG earnings weights. Appendix A1.1 provides the construction details.

the pandemic, and has remained elevated ever since.[8]

To understand why young graduates are worse off now, we examine transitions into and out of employment. This allows us to distinguish between two possible explanations: young graduates may have become more likely to lose their jobs, or they may face greater difficulties finding employment in the first place. We find that the deterioration is concentrated in their relative job-finding rates. Among unemployed college graduates, those aged 21–29 were 6 percentage points more likely to find a job the following month than those aged 30–55 in 2011–2015. This advantage fell to 2 points in 2016–2019 and disappeared by 2022–2024. Young graduates' job-finding rate changed little, while older graduates' rate improved. The gap in their employment-to-unemployment transition rates remained nearly unchanged. Together, these patterns suggest that the deterioration in young graduates' relative labor market outcomes reflects a growing disadvantage in finding employment rather than an increase in job losses.[9]

Several changes could contribute to these patterns. Beaudry et al. (2016) document a reversal in demand for cognitive tasks after 2000, even as the supply of college-educated workers continued to grow, while Cline and Kaymak (2025) discuss how slower growth in demand for college labor alongside rising college attainment could weaken graduates' job-finding prospects. More recently, generative AI has been associated with employment declines among young workers in exposed occupations, while remote work can reduce opportunities for junior workers (Brynjolfsson et al., 2025; Emanuel et al., 2026). However, the deterioration in young graduates' relative labor market outcomes began before the pandemic. One possible explanation is that the skills demanded by employers have changed in ways that favor experienced workers. While college provides technical training, skills such as coordinating with coworkers, managing clients, and making decisions may be acquired primarily through work experience. As employers place a greater premium on these skills, young college graduates may find it increasingly difficult to compete with older, more experienced workers. We therefore examine how employers' demand for soft and technical skills has changed over this period.

Advertised Skill Requirements. Lightcast job postings allow us to examine how employers' demand for soft and technical skills has changed over time. Figure 3 plots the average number of soft and technical skills required per job posting. Both increased over 2011–2024, with a larger proportional rise in soft skills. However, this aggregate increase could reflect either changes in skill requirements within occupations or a shift in job postings towards occupations that already require more skills. If employers are increasingly demanding soft skills even within existing occupations, young graduates may face greater difficulties finding employment across a broad range of jobs, rather than only in occupations that have traditionally required

[8] The share of young graduates with STEM bachelor's fields increased over 2011–2024, while unemployment rose relative to the overall rate within most of the broad fields shown, including STEM (Appendix Figures A10 and A11). The deterioration therefore does not appear to reflect only a shift away from STEM.

[9] Appendix A5 reports the matched CPS transitions and sample construction. Direct transitions from outside the labor force into employment increased for young graduates, so the loss of an advantage is specific to job finding from unemployment. Cline and Kaymak (2025) document a similar erosion when comparing young college and high-school graduates.

more soft skills.

To distinguish between these two possibilities, Table 2 examines changes in skill requirements within occupations, controlling for occupation fixed effects, seasonality, and average posting length. The estimates imply an increase of about 16% in soft-skill requirements and 6% in technical requirements over 2011–2024. Thus, even within the same occupation, employers are increasingly asking workers to possess a wider range of skills, with soft-skill requirements growing more than twice as fast as technical requirements.

Moreover, the increase in soft-skill requirements is widespread across occupations. Figure 4 compares requirements in six-digit SOC occupations observed in 2011–2015 with those in 2016–2024. Soft-skill requirements increased in 88% of occupations, representing 90% of postings in the initial period. Technical requirements increased in 85% of occupations. This suggests that rising soft-skill demand is not confined to a few occupations, but represents a broader change in the skills employers require. Workers therefore face higher requirements across much of the occupational distribution, rather than being concentrated in a few, traditionally soft-skill-intensive occupations.[10]

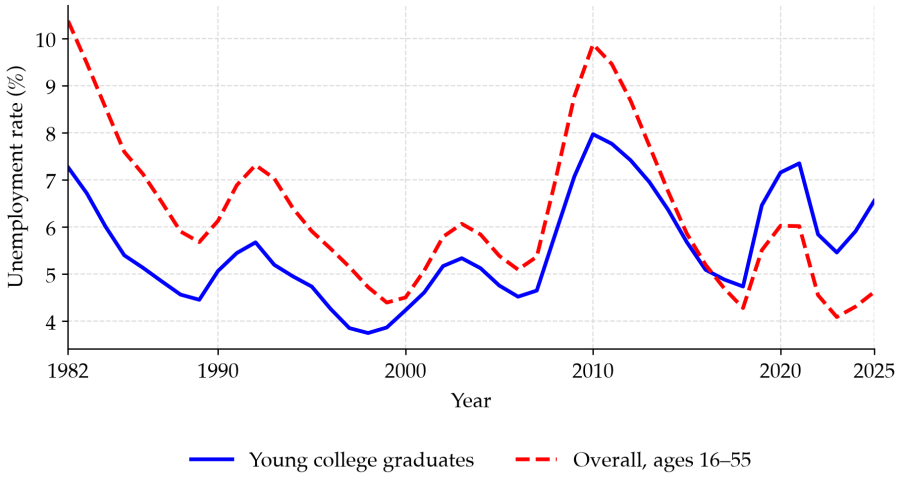
Skill Intensity. Higher skill counts could reflect a longer list of requirements without a change in the duties of the job. For example, employers may increasingly list generic requirements such as communication or teamwork without fundamentally changing what workers are expected to do. To understand whether the increase in soft-skill demand reflects more substantive changes in job requirements, we turn to the text-based classification described in Section I. This measure distinguishes generic mentions of soft skills from duties that require their substantial use, such as managing a team or coordinating across departments. It also allows a posting to require high intensity in both soft and technical skills.

Figure 5, Panel A, shows that the share of postings rated high in soft-skill intensity rises from 54% in 2012 to 61% in 2025. In contrast, the share rated high in technical-skill intensity changes very little, from 51% to 52%. These series hold the composition of 3-digit SOC occupation groups fixed, suggesting that the increase is not simply driven by a shift towards more soft-skill-intensive occupations. Together with the increase in skill counts documented earlier, this suggests that employers are not only listing more soft skills, but are increasingly requiring workers to perform tasks that involve substantial use of these skills.

Importantly, the increase in soft-skill intensity appears to be complementary to technical skills. Panel B shows that the share of postings rated high in both soft and technical intensity rises from 31% to 37%. Most of the increase in the soft-high share therefore occurs in postings that also require high technical intensity. At the same time, the share of postings requiring high technical but low soft-skill intensity declines. Among postings with high technical intensity, the share also requiring high soft-skill intensity rises by 9 percentage points. These patterns suggest that technical skills alone may increasingly be insufficient to meet employers' requirements. Instead, employers are seeking workers who can combine technical expertise with strong soft skills. This is particularly relevant for young college graduates, who may have

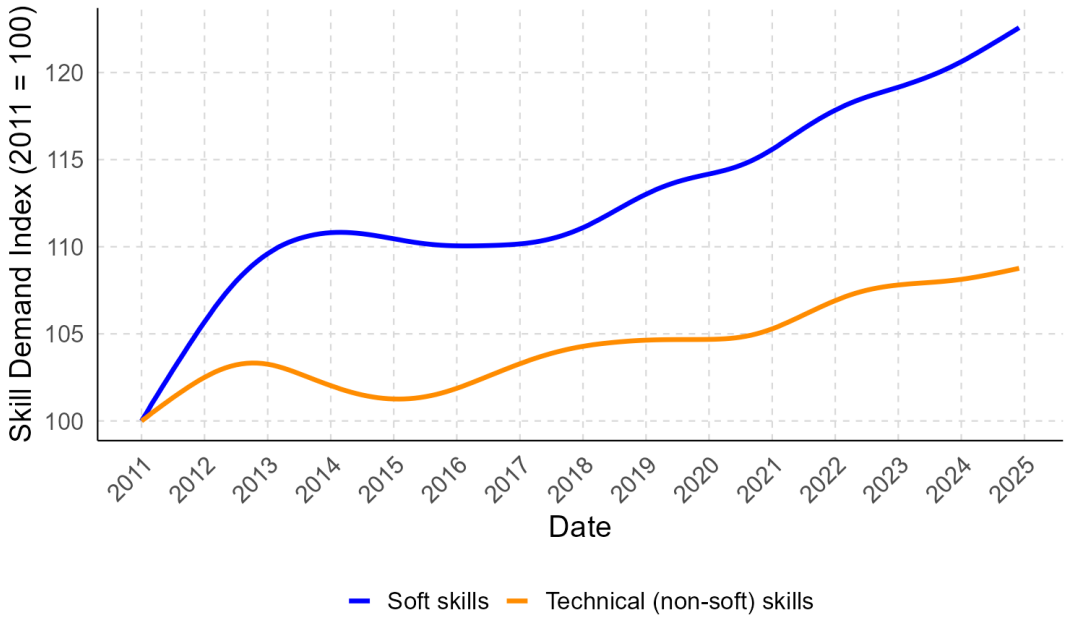
[10] Appendix Figure A3 compares changes in both skill categories and shows the occupational skill map in each period.

Figure 2: Young College Graduates' Unemployment Rate



Notes: Annual CPS unemployment rates, 1982–2025. Young workers are aged 16–25; the overall benchmark covers ages 16–55. College graduates hold a bachelor's degree or higher. The panel uses final CPS survey weights, includes the self-employed in the labor force. Rates are smoothed with a centered three-year moving average; endpoints show the unsmoothed annual rate.

Figure 3: Time Series For Skill Demand



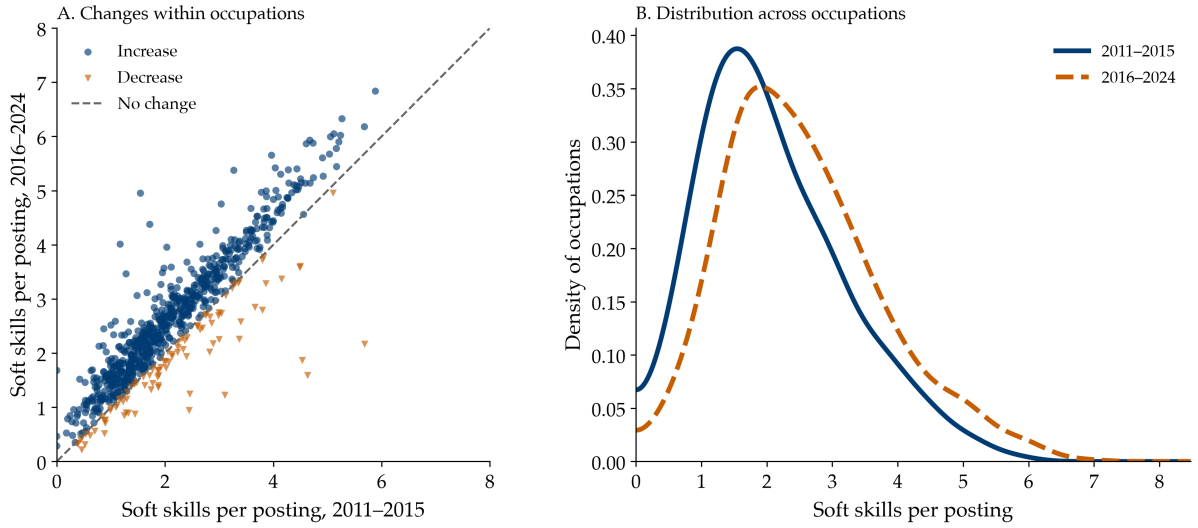
Notes: This figure shows changes in average skill counts per posting in Lightcast from 2011–2024. Soft skills follow the taxonomy rules in Appendix A1.2; technical skills comprise the remaining labels, including certifications. Monthly averages are total skill mentions divided by total postings, calculated separately for each category. Each series is smoothed using an HP filter with $\lambda = 14,400$ and then normalized to 100 in January 2011. These series use skill-label counts; the separate LLM intensity measure rates the requirements described in full advertisements.

Table 2: Rising Demand For Soft Skills

Dependent Variable:	Log(All Skills)	Log(Soft Skills)	Log(Technical Skills)
% change per year	0.60***	1.20***	0.48***
	(0.12)	(0.24)	(0.12)
Fixed-Effects:			
Occupation	Yes	Yes	Yes
Month	Yes	Yes	Yes

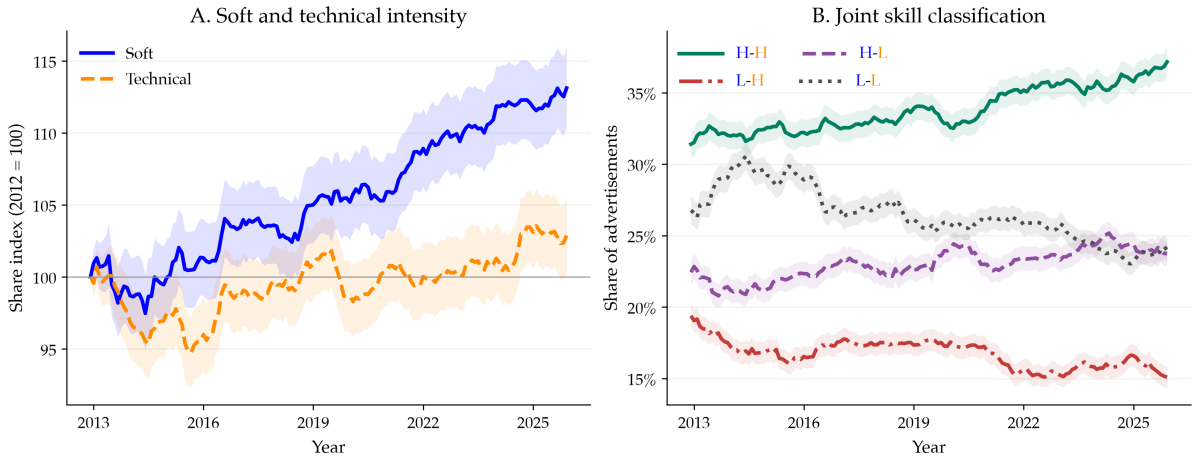
Notes: This table shows the slope of a linear time trend for skill demand after controlling for occupation and month fixed effects (Lightcast, 2011-2024). The unit of observation is an occupation × month × year group, where occupations are classified at the 6 digit SOC code level. Soft (technical) skill demand is calculated as the average number of soft (technical) skills required in a job posting in a given occupation × month × year group. All Skills = Soft Skills + Technical Skills. Coefficients are multiplied by 12 to reflect annual changes in skill demand. We also control for the average number of words per job posting within an occupation × month × year group. Each observation is weighted by the number of job postings within the group. Standard errors are clustered at the occupation level. $* \Rightarrow p < 0.1, ** \Rightarrow p < 0.05, *** \Rightarrow p < 0.01$.

Figure 4: The Shift in Occupational Skill Requirements



Notes: The sample consists of all six-digit SOC occupations observed in both 2011–2015 and 2016–2024 in the Lightcast data. Panel A plots each occupation's average soft-skill requirements in 2011–2015 against those in 2016–2024. Blue circles indicate increases and orange triangles indicate decreases; the dashed diagonal marks no change. Each point represents one occupation. Counts are total skill mentions divided by total postings within occupation and period. Panel B shows the distribution of average soft-skill requirements across the same occupations, giving each occupation equal weight. The solid dark-blue curve denotes 2011–2015 and the dashed orange curve denotes 2016–2024. Counts use the taxonomy rules in Appendix A1.2.

Figure 5: Skill Intensity in Job Postings



Notes: Panel A shows the soft-high and technical-high shares, each indexed to 100 for calendar year 2012. Panel B shows the four joint classification shares in levels. In its legend, the first letter is blue and denotes soft skills; the second is orange and denotes technical skills. H means high and L low. Separate line colors and patterns identify the four combinations, whose shares sum to one in each window. Both panels pool postings over rolling twelve-month windows and use sampling weights within each of 85 three-digit SOC groups; group shares are held fixed at their mean annual weighted shares over 2012–2025. The common sample contains 99,603 postings with both ratings. Shading shows approximate pointwise 95% confidence intervals.

the necessary technical training but lack the work experience needed to develop some of these additional capabilities.

The increase in soft-skill requirements is not limited to jobs requiring substantial prior work experience. We examine a separate sample of full-time postings requiring a bachelor's degree or higher and 0–2 years of experience. Holding occupational composition fixed, average soft-skill counts per posting rise from 3.61 to 4.72 over 2011–2024, an increase of 30.9%. For senior postings requiring at least six years of experience, the corresponding increase is 23.6%. Entry-level technical requirements also rise by 26.2%; the detailed comparison, including mid-level jobs and technical skills, remains in Appendix Figure A13. Thus, even jobs explicitly open to workers with little experience increasingly require soft skills. If some of these skills are primarily acquired on the job, recent graduates may find it increasingly difficult to meet employers' requirements despite having the necessary technical training.

Skill Requirements and the Young–Older Wage Gap. The evidence so far suggests that employers increasingly demand soft skills, including in jobs open to workers with limited experience. If these skills are primarily acquired through work experience, older workers should have an advantage over recent college graduates in occupations that require more soft skills. This advantage may be reflected not only in hiring decisions but also in wages. We therefore examine whether the wage gap between young and older college graduates is larger in occupations with higher soft-skill requirements.

We assign average soft and technical skill counts in each occupation and month to CPS respondents using their occupation codes and interview month.[11] We then relate wages to

[11] Deming and Kahn (2018) show that advertised cognitive and social skill requirements predict pay and firm

both requirements and interact each measure with an indicator for young workers:

$$\log w_{iot} = \beta_s \, \text{Soft}_{ot} + \beta_\tau \, \text{Tech}_{ot} + \gamma \, \text{Young}_i \tag{1}$$

$$+ \delta_s \, (\text{Young}_i \times \text{Soft}_{ot}) + \delta_\tau \, (\text{Young}_i \times \text{Tech}_{ot}) \tag{2}$$

$$+ X_i' \Theta + \lambda_t + \varepsilon_{iot}, \tag{3}$$

Here i denotes respondents, o occupations, and t months. Soft_{ot} and Tech_{ot} are the average skill counts per posting, and Young_i identifies workers aged 25 or younger. We estimate the regression for college-educated workers using CPS MORG earnings from 2011–2024, controlling for sex, race, education, and time fixed effects. The coefficients of interest are δ_s and δ_τ, which capture how the wage gap between young and older college graduates varies with the soft- and technical-skill requirements of their occupations.

Table 3 shows that the wage gap between young and older college graduates is larger in occupations requiring more soft skills. One additional soft skill per job posting is associated with approximately 0.03 log points lower wages for young graduates relative to older graduates. This is consistent with our hypothesis that older workers have an advantage in occupations where skills acquired through work experience are more valuable.

Technical requirements show a different pattern in the baseline regression. One additional technical skill per posting is associated with a 0.01-log-point increase in young college graduates' wages relative to older graduates' wages. This pattern fits a role for college training in preparing young graduates to meet prevalent technical requirements.[12] The soft-skill wage gap persists when we account for differences in average pay across occupations. With occupation fixed effects, the college soft interaction remains about −0.03 log points. Additional specifications using entry-level postings and within-occupation changes are reported in Appendix A4.6.

These results suggest that young college graduates face lower wages in occupations where employers require more soft skills. One possible explanation is that these skills are accumulated through work experience, giving older workers an advantage over recent graduates. If the demand for such skills has increased over time, we would also expect the wage premium associated with work experience to have risen. We examine this prediction next.

The Experience Premium. We next examine whether the wage advantage of experience has grown for more recent cohorts of college graduates. Figure 6 plots estimated wage–experience profiles from the CPS MORG for cohorts reaching age 22 between 1999 and 2015. Later cohorts have steeper profiles. At ten years of potential experience, the estimated wage premium relative to zero experience rises from 31% for the 1999 cohort to 65% for the 2015 cohort. The larger premium suggests that work experience has become more important for college graduates' earnings. As soft skills become more valuable, workers who have had more time to develop them may command a larger wage premium over recent graduates. While these pat-

performance, even within narrowly defined occupations.

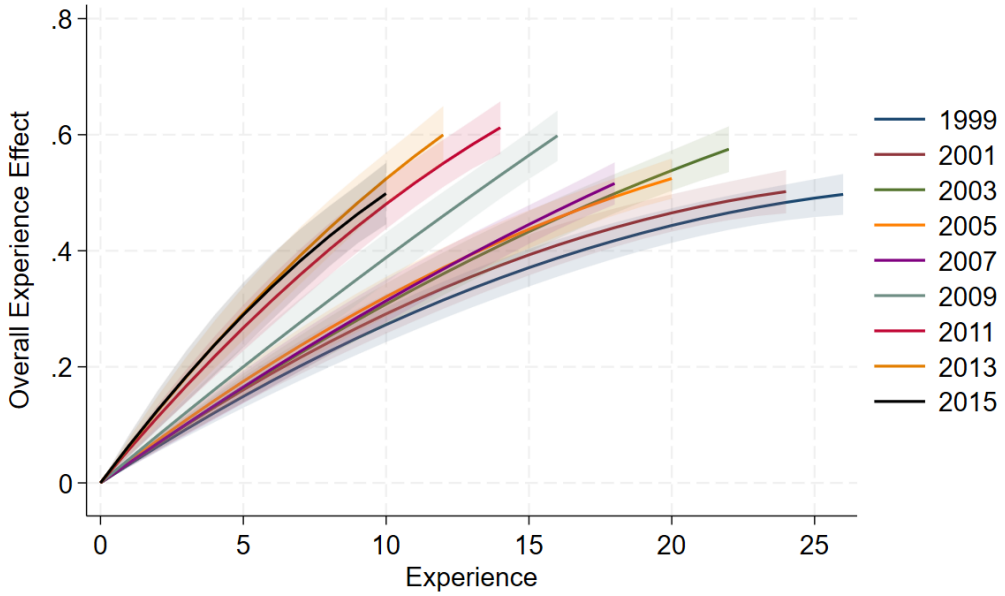
[12] Deming and Noray (2020) document a related pattern: the earnings premium for technology-intensive college degrees is high early in workers' careers and declines with experience as their skills become obsolete.

Table 3: Young College Graduates Earn Less in Occupations Requiring More Soft Skills

Dependent Variable: Log(Hourly Wage)	Skill Composition	
	Time-Varying	Fixed
	(1)	(2)
Young (γ)	-0.3724^{***}	-0.3462^{***}
	(0.0405)	(0.0442)
Soft skills (β_s)	0.0266^{*}	0.0278^{*}
	(0.0151)	(0.0168)
Technical skills (β_τ)	0.0296^{***}	0.0314^{***}
	(0.0030)	(0.0033)
Young × Soft (δ_s)	-0.0300^{**}	-0.0329^{**}
	(0.0142)	(0.0158)
Young × Technical (δ_τ)	0.0093^{***}	0.0078^{**}
	(0.0033)	(0.0032)
Time fixed effects	Yes	Yes
Demographic controls	Yes	Yes
Observations	572,091	572,091
R^2	0.161	0.166
Mean log hourly wage	3.32	3.32
Share young	0.090	0.090
Mean soft skills per posting	3.09 (1.35)	3.16 (1.27)
Mean technical skills per posting	10.55 (4.32)	10.71 (4.13)

Notes: This table shows how the returns to occupations' soft and technical skill requirements differ between young and older college-educated workers. Sources: CPS Merged Outgoing Rotation Groups and Lightcast job postings, 2011–2024. The sample includes wage-and-salary workers aged 54 or younger with a bachelor's or advanced degree and positive real hourly wages, in occupations that can be linked to Lightcast postings. Hourly wages are reported hourly wages or, if those are missing, weekly earnings divided by usual weekly hours. They are deflated by the PCE price index. Young = 1 if age ≤ 25 and 0 if age $\in [26, 54]$. The soft (technical) skills variable is the average number of soft (technical, i.e. non-soft) skills required per Lightcast job posting. In column (1), it is computed separately for each occupation and month. In column (2), it is computed once per occupation by pooling all postings from 2011 to 2024, so each occupation's skill requirements are held fixed over time. Occupations are harmonized OCC2010 groups linked to SOC codes. Demographic controls are indicators for sex, race and degree (bachelor's or advanced). Regressions are weighted by CPS final weights adjusted to hold demographic composition constant over time. Standard errors in parentheses are clustered at the occupation level. Means, shares and standard deviations in the bottom panel are weighted and computed over the regression sample. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 6: Wage–Experience Profiles of College Graduates by Cohort



Notes: Source: CPS MORG. The sample includes workers aged 16–55 with a bachelor's degree or higher, positive real hourly earnings, and positive earnings weights; allocated wages are excluded. Cohorts are defined by the year respondents reach age 22. Potential experience is age minus years of schooling minus six. Separate earnings-weighted regressions for each cohort relate log real hourly earnings to experience, experience squared, and indicators for education, sex, race, occupation, and industry. Each curve plots $\widehat{\beta}_{1c}x + \widehat{\beta}_{2c}x^2$, the fitted log wage difference relative to zero experience, where x denotes potential experience. Shading shows pointwise 95% confidence intervals calculated using the delta method. Profiles are displayed through 2025 $-$ c years for cohort c.

terns do not directly establish that soft-skill accumulation drives the increase in the experience premium, they are consistent with our proposed mechanism.[13]

In summary, we document several novel facts about the worsening labor market dynamics for young workers, and link these to the increase in demand for soft skills. While young college graduates historically had an unemployment rate below the overall rate, this pattern has reversed since 2017. The advantage in job-finding that they once had over older graduates has also disappeared, while relative separation rates have not changed much. We document that over the same period, employers have increasingly demanded soft skills, both across and within occupations, including in jobs requiring little prior work experience. We also find that young graduates earn relatively less in occupations requiring more soft skills, and that the wage premium associated with experience has increased across cohorts. These patterns are consistent with our hypothesis that rising demand for skills accumulated on the job has placed young college graduates at a disadvantage relative to older, more experienced workers.

These findings suggest that young workers are worse-off relative to older, experienced workers, especially in occupations requiring more soft skills. Given that most of the occupations have seen an increase in soft skill requirements, employers seem to be increasingly favoring experienced workers over recent graduates, who enter with technical training but

[13] Entry earnings also matter for lifetime outcomes. Guvenen et al. (2022) show that differences in median lifetime earnings across cohorts largely reflect earnings differences already present at labor-market entry.

have had less time to acquire soft skills on the job. This may result in a higher unemployment rate for young college graduates relative to the overall rate. However, the resulting changes in vacancy creation and job opportunities depend on how firms respond to this shift in demand. Young workers can accumulate skills through employment, which may improve their future job prospects. To understand how these forces interact, we build a general equilibrium search-and-matching model with heterogeneous workers and firms and on-the-job skill accumulation. The model allows us to examine how rising skill demand changes the distribution of vacancies across firms, and whether this reduces job-finding opportunities for young, inexperienced workers. We then quantify how much of the increase in young graduates' relative unemployment can be explained by this mechanism.

III Model

We will now build a general equilibrium, search and matching model of the US labor market, using the model developed by Lise and Robin (2017) as a baseline. The model contains heterogeneous workers and firms, search is random, and workers can accumulate skills on-the-job. The goal of the model is to quantify the contribution of rising soft skill demand to the worsening of labor market outcomes of young college graduates.

III.A Workers, Firms, and States

Time is discrete. The economy contains a continuum of workers identified by their skill level $x_t \in \mathcal{X} = \{\underline{x}, \underline{x} + \Delta, \ldots, \bar{x}\}$, and a continuum of firms characterized by productivity $y \in \mathcal{Y} = [0, 1]$. A worker's skill level evolves over her lifetime. While employed, in each period a worker climbs one skill level with probability π and remains at her current skill level with probability $1 - \pi$. Workers exit the labor force at rate ξ and are replaced one for one by entrants, who enter unemployed with a skill level drawn from a fixed Beta distribution ψ on $\mathcal{X}$:

$$n_t(x_t) = \xi\, \psi(x_t), \qquad \int_{\mathcal{X}} \psi(x) dx = 1. \tag{4}$$

The distribution of productivities across firms is uniform, and a firm's productivity is fixed over time. Both workers and firms are risk neutral. The aggregate state of the economy is given by z_t. At the beginning of each period t, the aggregate state evolves according to a Markov transition kernel $\Pi_z(z' \mid z)$.

III.B Production and Flow Value of Unemployment

The output produced by a match between a worker with skills x_t and a firm with productivity y at date t is given by a second-order Taylor polynomial in the worker and firm type, scaled up by the level of aggregate shock z_t.

$$p_t(x_t, y, z) = z_t(a_1 + a_2 x_t + a_3 y + a_4 x_t^2 + a_5 y^2 + a_6 x_t y) \tag{5}$$

This quadratic specification allows for the output to be non-monotonic in either the firm's or the worker's type. For example, if $a_5 < 0$, then it is possible that for some worker types

x_t, moving to a high-type firm $y' > y$ leads to a lower level of output $p_t(x_t, y', z) < p_t(x_t, y, z)$. Similarly, the production function allows for complementarity between the worker and firm types (if $a_6 \neq 0$). Therefore, this specification will be an important driver of sorting between firms and workers, which will become apparent later.

For an unmatched worker, the flow value of unemployment is given as a fixed proportion of the output she can produce at her best possible firm, in the absence of any aggregate shocks:

$$b_t(x_t) = 0.7 \times p_t(x_t, y^*(x_t, 1), 1) \tag{6}$$

where the 0.7 factor is taken from Hall and Milgrom (2008), and $y^*(x_t, 1) = \arg\max_y p_t(x_t, y, 1)$ is the firm type that maximizes a worker's output in the absence of aggregate shocks ($z = 1$).

III.C Labor Markets

Consider the labor market at the beginning of period t. Let $u_t(x_t)$ denote the mass of unemployed workers with skills x_t, and let $h_t(x_t, y)$ denote the mass of employed workers with skills x_t currently matched with firm y. The total mass of workers is normalized to one:

$$\int_{\mathcal{X}} u_t(x_t)\, dx + \int_{\mathcal{X}} \int_{\mathcal{Y}} h_t(x_t, y)\, dy\, dx = 1. \tag{7}$$

The timeline of the labor market is as follows: The period t starts with a change in the aggregate state from z_{t-1} to z_t. Separations between workers and firms happen, since some matches that were viable in period $t-1$ (i.e., generated a positive surplus) are not viable anymore. The resulting stock of unemployed agents is denoted by $u_t^+(x_t)$ and the remaining employed workers are denoted by $h_t^+(x_t, y)$. The effective labor force (the mass of workers searching for jobs) is given by

$$L_t = \int_{\mathcal{X}} u_t^+(x_t) dx + s \int_{\mathcal{X}} \int_{\mathcal{Y}} h_t^+(x_t, y)\, dy\, dx, \tag{8}$$

where $s \in [0, 1]$ is the relative search efficiency of employed workers (the search effort of unemployed workers is normalized to 1).

During the search phase, firms of type y post vacancies $v_t(y)$. The total number of vacancies in the labor market is given by $V_t = \int_{\mathcal{Y}} v_t(y)\, dy$. Matches form according to

$$\mathcal{M}_t(L, V) = \min\{\alpha_M L^{\omega} V^{1-\omega}, V_t, L_t\}, \tag{9}$$

where we let $\omega = 1/2$, and market tightness is defined as $\theta_t = V_t / L_t$. The job finding rate is

$$\lambda_{w,t} = \frac{\mathcal{M}(L_t, V_t)}{L_t} = \alpha_M \theta_t^{1/2}, \tag{10}$$

and the job filling rate is

$$\lambda_{f,t} = \frac{\mathcal{M}(L_t, V_t)}{V_t} = \alpha_M \theta_t^{-1/2}. \tag{11}$$

Search is random. Each period, an unemployed worker receives a job offer with probability $\lambda_{w,t}$ and an employed worker receives a job offer with probability $s\lambda_{w,t}$. Similarly, a firm posting a vacancy gets in contact with a worker with probability $\lambda_{f,t}$. Production, skill accumulation, and exit and entry follow the search phase, as described in Section III.F.

III.D Values and Surplus

III.D.1 Unemployment

Let $B_t(x_t)$ denote the value of unemployment for a worker with skills x_t. If a worker remains unemployed throughout period t, she receives the value $b_t(x_t)$ (as defined in Equation 6). In period $t+1$, this worker can receive a job offer with probability $\lambda_{w,t+1}$. We assume that the worker has zero bargaining power, so any firm y matched with this worker will offer her reservation value $B_{t+1}(x_{t+1})$. If the worker does not receive a job offer (with probability $1-\lambda_{w,t+1}$), she remains unemployed and receives her value of unemployment $B_{t+1}(x_{t+1})$. Moreover, the worker exits the labor force with probability ξ, which discounts the continuation value. Therefore, the expression for $B_t(x_t)$ is given as follows:

$$\begin{aligned} B_t(x_t) &= b_t(x_t) + \beta(1-\xi)\mathbb{E}_t\left[(1-\lambda_{w,t+1})B_{t+1}(x_{t+1}) + \lambda_{w,t+1}\int_{\mathcal{Y}} \frac{v_{t+1}(y)}{V_{t+1}} B_{t+1}(x_{t+1})\,dy\right] \\ &= b_t(x_t) + \beta(1-\xi)\,\mathbb{E}_t\Big[B_{t+1}(x_{t+1})\Big] \end{aligned} \tag{12}$$

where β is the discount factor and $\mathbb{E}_t$ is the expectation taken at time t with respect to the future aggregate states z.

III.D.2 Surplus

Let $P_t(x_t, y, z)$ denote the joint value of a match between a worker x_t and a type-y firm. Define match surplus as

$$S_t(x_t, y, z) = P_t(x_t, y, z) - B_t(x_t). \tag{13}$$

A match between a worker x_t and firm y produces output $p_t(x_t, y, z)$ in period t. In period $t+1$, after the worker's skill appreciation and the realization of the aggregate shock z_{t+1}, the worker and firm assess whether to continue with the match or not. Since matches are efficient, a match is destroyed if $P_{t+1}(x_{t+1}, y, z) < B_{t+1}(x_{t+1}) \Leftrightarrow S_{t+1}(x_{t+1}, y, z) < 0$,[14] otherwise the match remains viable (i.e., if $P_{t+1}(x_{t+1}, y, z) \geq B_{t+1}(x_{t+1}) \Leftrightarrow S_{t+1}(x_{t+1}, y, z) \geq 0$). Matches can also be destroyed with an exogenous probability δ in each period.[15]

If a match remains viable in period $t+1$, the worker draws a job offer with probability $s\lambda_{w,t+1}$, and the probability that this job offer comes from a firm of type y' is $v_{t+1}(y')/V_{t+1}$. Whether the worker accepts or rejects the offer depends on the value offered by her current firm $W_{t+1}(x_{t+1}, y, z, \sigma_{t+1})$ and the competing firm $W_{t+1}(x_{t+1}, y', z, \sigma_{t+1})$, where $\sigma_t \in [0,1]$ is the share of match surplus that the worker gets. We use the sequential auction framework of Postel-Vinay and Robin (2002) to model the competition between firms and determine the wages offered to the worker.

Suppose the worker's current value at firm y is $W_t(x_t, y, z, \sigma_t) \leq P_t(x_t, y, z)$. If $P_{t+1}(x_{t+1}, y', z) > P_{t+1}(x_{t+1}, y, z) \Leftrightarrow S_{t+1}(x_{t+1}, y', z) > S_{t+1}(x_{t+1}, y, z)$, then the two firms compete with each other by

[14] The continuation value of a match post-separation is $B_{t+1}(x_{t+1})$, as the firm receives a value of 0.

[15] Abusing the notation slightly, we remove the time subscript on the aggregate shock z. It should be interpreted as follows: If the value function is for the period t, for example $P_t(x_t, y, z)$, then $z = z_t$. If the value function is for period $t+1$, for example $P_{t+1}(x_{t+1}, y, z)$ then $z = z_{t+1}$. Note that a firm's type y remains fixed across periods.

raising their value offers, until the value offer reaches $W_{t+1}(x_{t+1}, y', z, \sigma_{t+1}) = W_{t+1}(x_{t+1}, y, z, \sigma_{t+1}) = P_{t+1}(x_{t+1}, y, z)$ and the worker switches to the new firm y'.[16] On the other hand, if $P_{t+1}(x_{t+1}, y', z) \leq P_{t+1}(x_{t+1}, y, z) \Leftrightarrow S_{t+1}(x_{t+1}, y', z) \leq S_{t+1}(x_{t+1}, y, z)$, then the worker stays at her current firm, but she will receive a new value $W_{t+1}(x_{t+1}, y, z, \sigma_{t+1}) = \max[W_t(x_{t+1}, y, z, \sigma_t), P_{t+1}(x_{t+1}, y', z)]$.[17]

Future output from a worker-firm match also needs to account for the fact that workers can accumulate skills on-the-job: With probability π the worker's skills rise to $x_{t+1} = x_t + \Delta$, and with probability $1 - \pi$ she remains at $x_{t+1} = x_t$. For any function $f(x_t)$, define the expectation over an employed worker's skill level in the next period, given her current skills $x_t < \bar{x}$, as:[18]

$$\mathbb{E}_\pi f(x_{t+1}) \equiv \pi f(x_t + \Delta) + (1 - \pi) f(x_t). \tag{14}$$

Skill accumulation happens after current production and before next period's separation and poaching stages. Therefore we get the following expression for the value of a match:

$$\begin{aligned}
P_t(x_t, y, z) &= p_t(x_t, y, z) \\
&\quad + \beta(1-\xi)\mathbb{E}_t\mathbb{E}_\pi\Big[\Big(\delta + (1-\delta)\mathbf{1}\{P_{t+1}(x_{t+1}, y, z) < B_{t+1}(x_{t+1})\}\Big)B_{t+1}(x_{t+1}) \\
&\qquad + (1-\delta)\mathbf{1}\{P_{t+1}(x_{t+1}, y, z) \geq B_{t+1}(x_{t+1})\}\Big((1 - s\lambda_{w,t+1})P_{t+1}(x_{t+1}, y, z) \\
&\qquad\qquad + s\lambda_{w,t+1}\int_{\mathcal{Y}} P_{t+1}(x_{t+1}, y, z)\frac{v_{t+1}(y')}{V_{t+1}}\,dy'\Big)\Big] \\
&= p_t(x_t, y, z) \\
&\quad + \beta(1-\xi)\mathbb{E}_t\mathbb{E}_\pi\Big[\Big(1 - (1-\delta)\mathbf{1}\{S_{t+1}(x_{t+1}, y, z) \geq 0\}\Big)B_{t+1}(x_{t+1}) \\
&\qquad + (1-\delta)\mathbf{1}\{S_{t+1}(x_{t+1}, y, z) \geq 0\}P_{t+1}(x_{t+1}, y, z)\Big].
\end{aligned} \tag{15}$$

Note that $P_t(x_t, y, z)$ is independent of the identity of the competing firm y'. This is because if the worker remains at firm y in period $t+1$, the value $P_{t+1}(x_{t+1}, y, z)$ is not affected by the competing offer (only the share received by the worker changes). If the worker switches, the firm y receives 0, while the worker receives a value $W_{t+1}(x_{t+1}, y', z, \sigma_{t+1}) = P_{t+1}(x_{t+1}, y, z)$.

Combining equations (12), (13), and (15), we obtain the following expression for the match

[16] At this point, firm y cannot raise its value offer anymore, but firm y' can still raise its offer by an ϵ amount and poach the worker, so we break ties in favour of the firm which has a higher surplus with the worker.

[17] There is a third case, where the worker receives a very low value offer ($W_t(x_{t+1}, y, z, \sigma_t) \geq P_{t+1}(x_{t+1}, y', z)$). In this case, the worker rejects the offer and her value remains unchanged.

[18] The skill level $x = \bar{x}$ is absorbing. No skill accumulation happens if a worker is already at the highest skill level.

surplus:

$$
\begin{aligned}
S_t(x_t, y, z) &= p_t(x_t, y, z) - b_t(x_t) \\
&\quad + \beta(1-\xi)\,\mathbb{E}_t\Big[\pi\, B_{t+1}(x_t + \Delta) + (1-\pi)\, B_{t+1}(x_t) - B_{t+1}(x_t)\Big] \\
&\quad + \beta(1-\xi)(1-\delta)\,\mathbb{E}_t\Big[\pi\, \mathbf{1}\{S_{t+1}(x_t+\Delta, y, z) \geq 0\}\Big(P_{t+1}(x_t+\Delta, y, z) - B_{t+1}(x_t+\Delta)\Big) \\
&\qquad\qquad + (1-\pi)\,\mathbf{1}\{S_{t+1}(x_t, y, z) \geq 0\}\Big(P_{t+1}(x_t, y, z) - B_{t+1}(x_t)\Big)\Big] \\
&= p_t(x_t, y, z) - b_t(x_t) \\
&\quad + \beta(1-\xi)\,\mathbb{E}_t\Big[\mathbb{E}_\pi\Big((1-\delta)\,\mathbf{1}\{S_{t+1}(x_{t+1}, y, z) \geq 0\}\Big(P_{t+1}(x_{t+1}, y, z) - B_{t+1}(x_{t+1})\Big)\Big)\Big] \\
&\quad + \beta(1-\xi)\,\mathbb{E}_t\Big[\mathbb{E}_\pi B_{t+1}(x_{t+1}) - B_{t+1}(x_t)\Big]. \qquad (16)
\end{aligned}
$$

The final term in Equation 16 accounts for the change in the value of unemployment for a worker when she is employed rather than unemployed. Proposition 1 in Lise and Robin (2017) states that the match surplus depends on time only through the aggregate productivity shock z_t, and it does not depend on the distributions of vacancies, unemployed workers, or worker-firm matches in our model without skill accumulation. An analogous result holds for our model with skill accumulation.

III.E Vacancy Posting

The cost of posting $v_t(y)$ vacancies is increasing and convex, and there is no fixed cost required for the creation of vacancies:

$$
c(v_t(y)) = \kappa_0 \frac{v_t(y)^{1+\kappa_1}}{1+\kappa_1}, \qquad \kappa_0 > 0,\ \kappa_1 > 0. \qquad (17)
$$

Let $J_t(y)$ denote the expected value of filling a vacancy. If a vacancy does not lead to a viable match, the firm receives 0 value. If a match is formed, the value to the firm depends on the surplus generated from the match:

$$
\begin{aligned}
J_t(y) &= \int_{\mathcal{X}} \frac{u_t^+(x_t)}{L_t} \max\{S_t(x_t, y, z), 0\}\, dx \\
&\quad + \int_{\mathcal{X}} \int_{\mathcal{Y}} \frac{s\, h_t^+(x_t, y')}{L_t} \max\{S_t(x_t, y, z) - S_t(x_t, y', z), 0\}\, dy'\, dx. \qquad (18)
\end{aligned}
$$

The firm's vacancy-posting condition is

$$
\kappa_0 v_t^{\kappa_1}(y) = \lambda_{f,t} J_t(y), \qquad (19)
$$

which gives us the number of vacancies created by each firm type y:

$$
v_t(y) = \left(\lambda_{f,t} \frac{J_t(y)}{\kappa_0}\right)^{\frac{1}{\kappa_1}}. \qquad (20)
$$

Substituting for the job filling rate and aggregating over firm types gives us the equilibrium tightness:

$$
\theta_t = \left(\frac{1}{L_t} \int_{\mathcal{Y}} \left(\frac{\alpha_M J_t(y)}{\kappa_0}\right)^{\frac{1}{\kappa_1}} dy\right)^{\frac{\kappa_1}{\kappa_1 + 0.5}}. \qquad (21)
$$

III.F Worker Flows

Each period has multiple stages. Separations and matching operate first, and depend on the surplus $S_t(x_t, y, z)$ evaluated at the skill level x_t a worker holds at the start of the period. Then, production takes place. After production, employed workers accumulate skills. Finally, workers exit the labor force at an exogenous rate ξ, and new workers of the same mass enter the labor force with skill levels drawn from ψ.

During the separations stage, some employed workers will join the pool of unemployed either due to the exogenous job destruction shock δ, or because their match is not viable anymore ($S_t(x_t, y, z) < 0$). The resulting stocks of unemployed and employed workers are:

$$u_t^+(x_t) = u_t(x_t) + \int_{\mathcal{Y}} \Big(\delta + (1-\delta)\mathbf{1}\{S_t(x_t, y, z) < 0\}\Big) h_t(x_t, y)\, dy \tag{22}$$

and

$$h_t^+(x_t, y) = (1-\delta)\mathbf{1}\{S_t(x_t, y, z) \geq 0\} h_t(x_t, y). \tag{23}$$

During the matching and poaching stage, unemployed workers receive a job offer from a firm y with probability $\lambda_{w,t}\frac{v_t(y)}{V_t}$ and form a match with the firm if it is viable ($S_t(x_t, y, z) \geq 0$). The stock of employed workers at a firm comes from three sources: (i) workers who were not poached by other firms, (ii) workers poached from another firm, and (iii) workers hired from unemployment. Because the skill transition has not yet occurred, the resulting distribution of unemployed and employed workers is not yet the distribution at the beginning of period $t+1$. We denote it by $u_{t+1/2}(x_t)$ and $h_{t+1/2}(x_t, y)$:

$$u_{t+1/2}(x_t) = u_t^+(x_t) - u_t^+(x_t)\lambda_{w,t} \int_{\mathcal{Y}} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x_t, y, z) \geq 0\}\, dy \tag{24}$$

and

$$\begin{aligned} h_{t+1/2}(x_t, y) = h_t^+(x_t, y)\Bigg[1 - \int_{\mathcal{Y}} s\lambda_{w,t} \frac{v_t(y')}{V_t} \mathbf{1}\{S_t(x_t, y, z) < S_t(x_t, y', z)\}\, dy'\Bigg] \\ + \int_{\mathcal{Y}} h_t^+(x_t, y') s\lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x_t, y, z) > S_t(x_t, y', z)\}\, dy' \\ + u_t^+(x_t)\lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x_t, y, z) \geq 0\}. \end{aligned} \tag{25}$$

Production takes place with the workers in $h_{t+1/2}(x_t, y)$. The skill transition is then applied: an employed worker climbs one skill level with probability π, and remains at her current skill level with probability $1-\pi$. Exit and entry follow: exit removes a fraction ξ of every skill level, and entrants replace that mass one for one, spread across skill levels according to ψ. Combining the two steps, the distribution of employed workers at the beginning of period $t+1$ is

$$h_{t+1}(x_{t+1}, y) = \begin{cases} (1-\xi)\Big[\pi\, h_{t+1/2}(x_{t+1} - \Delta, y) + (1-\pi)\, h_{t+1/2}(x_{t+1}, y)\Big], & \underline{x} < x_{t+1} < \bar{x}, \\ (1-\xi)(1-\pi)\, h_{t+1/2}(\underline{x}, y), & x_{t+1} = \underline{x}, \\ (1-\xi)\Big[h_{t+1/2}(\bar{x}, y) + \pi\, h_{t+1/2}(\bar{x} - \Delta, y)\Big], & x_{t+1} = \bar{x}, \end{cases} \tag{26}$$

where the third case in Equation 26 reflects that the top skill level is absorbing: a worker at the highest skill level cannot accumulate skills anymore. The distribution of unemployed workers is given as follows:

$$u_{t+1}(x_{t+1}) = (1-\xi)u_{t+1/2}(x_{t+1}) + \xi\,\psi(x_{t+1}). \tag{27}$$

Unemployed workers are unable to accumulate skills. A fraction ξ of them exit the labor force each period, and new workers enter as unemployed. Both steps conserve mass: (24)–(25) leave total mass at 1, applying the factor $1-\xi$ to every skill level removes a total mass of ξ, and adding entrants from distribution ψ back in (27) restores it.

III.G Aggregate Quantities

Aggregate output is

$$Y_t = \int_{\mathcal{X}} \int_{\mathcal{Y}} p_t(x_t, y, z) h_{t+1/2}(x_t, y)\, dy\, dx \tag{28}$$

and aggregate consumption is

$$C_t = Y_t + \int_{\mathcal{X}} b_t(x_t)\, u_{t+1/2}(x_t)\, dx - \int_{\mathcal{Y}} \kappa_0 \frac{v_t(y)^{1+\kappa_1}}{1+\kappa_1}\, dy. \tag{29}$$

III.H Equilibrium

Given an initial distribution (u_0, h_0) and aggregate shock Π_z, a dynamic equilibrium is a collection

$$\{B_t, P_t, S_t, J_t, v_t, u_t, h_t, u_{t+1/2}, h_{t+1/2}, Y_t, C_t, \theta_t\}_{t\geq 0} \tag{30}$$

such that:

1. B_t, P_t, and S_t satisfy the Bellman equations (12), (15), and (16);
2. J_t is consistent with expected gains from hiring and poaching, as in (18);
3. vacancy posting satisfies the firm's first-order condition (19);
4. market tightness θ_t is determined by the matching function and by search and vacancy-creation choices;
5. unemployment and employment distributions evolve according to the worker-flow equations and the state-transition kernels;
6. goods-market feasibility holds in every period.

III.I Wages

An interesting property of our model is that we do not need wages to determine labor market flows at any stage. All vacancy posting decisions and worker moves are decided based on the surplus of a match (Lise and Robin, 2017). However, we are also interested in tracking wages of workers, so we derive the expression for a worker's wages as described below.

Let σ_t be the share of surplus $S_t(x_t, y, z) = P_t(x_t, y, z) - B_t(x_t)$ that a worker x_t receives when matched with a firm y in period t. This share can depend on the history of past offers this worker has received. As in Lise and Postel-Vinay (2020), we assume that σ_t remains constant until the match dissolves or until an outside offer arrives and renegotiation happens:

$$\sigma_t(x_t, y, z) = \frac{W_t(x_t, y, z, \sigma_t) - B_t(x_t)}{P_t(x_t, y, z) - B_t(x_t)}. \tag{31}$$

Let $W_t(x_t, y, z, \sigma_t)$ denote the value a worker obtains from being employed, and let $w_t(x_t, y, z, \sigma_t)$ denote the wage that delivers this value. Then

$$\begin{aligned}
W_t(x_t, y, z, \sigma_t) = {} & w_t(x_t, y, z, \sigma_t) \\
& + \beta(1-\xi)\,\mathbb{E}_t\mathbb{E}_\pi\Big[\Big(\delta + (1-\delta)\mathbf{1}\{S_{t+1}(x_{t+1}, y, z) < 0\}\Big)B_{t+1}(x_{t+1})\Big] \\
& + \beta(1-\xi)(1-\delta)\,\mathbb{E}_t\mathbb{E}_\pi\Big[\mathbf{1}\{S_{t+1}(x_{t+1}, y, z) \geq 0\}\Big(1 - s\lambda_{w,t+1}\Big)W_{t+1}(x_{t+1}, y, z, \sigma_{t+1})\Big] \\
& + \beta(1-\xi)(1-\delta)\,\mathbb{E}_t\mathbb{E}_\pi\Big[\mathbf{1}\{S_{t+1}(x_{t+1}, y, z) \geq 0\}\, s\lambda_{w,t+1} \\
& \quad \times \int_{\mathcal{Y}} \min\Big\{P_{t+1}(x_{t+1}, y, z), \max\{W_{t+1}(x_{t+1}, y, z, \sigma_{t+1}), P_{t+1}(x_{t+1}, y', z)\}\Big\}\frac{v_{t+1}(y')}{V_{t+1}}\,dy'\Big].
\end{aligned} \tag{32}$$

Since skill accumulation happens before the separation and poaching stages, the expectation $\mathbb{E}_\pi$ (defined in (14)) wraps each continuation value, as in (15). The first term denotes current period wages. The second term considers the situation where the worker becomes unemployed next period.[19] The third term refers to the case when the match is not destroyed and the worker does not receive a job offer from a competing firm. In this case, she receives the same share of surplus σ_t that she received in the previous period.

The final term provides the worker's value if an outside offer arrives. The actual value received by the worker depends on multiple cases: (i) If the outside offer is low enough, i.e. $W_{t+1}(x_{t+1}, y, z, \sigma_{t+1}) \geq P_{t+1}(x_{t+1}, y', z)$, where $P_{t+1}(x_{t+1}, y', z)$ is the maximum value that firm y' can offer, then the worker stays at her current firm and receives $W_{t+1}(x_{t+1}, y, z, \sigma_{t+1})$, (ii) If the outside offer is large enough, i.e. $P_{t+1}(x_{t+1}, y', z) > P_{t+1}(x_{t+1}, y, z)$, then the worker switches to firm y' and receives a value of $P_{t+1}(x_{t+1}, y, z)$, and (iii) in the intermediate case when $P_{t+1}(x_{t+1}, y, z) \geq P_{t+1}(x_{t+1}, y', z) > W_{t+1}(x_{t+1}, y, z, \sigma_{t+1})$, the worker renegotiates her value up to $P_{t+1}(x_{t+1}, y', z)$ and stays at her current firm y.

The expression for $w_t(x_t, y, z, \sigma_t)$ can then be written by rearranging Equation (32), using Equation (31) to obtain $W_{t+1}(x_{t+1}, y, z, \sigma_{t+1})$ in terms of $P_{t+1}(x_{t+1}, y, z)$ and $B_{t+1}(x_{t+1})$, and then substituting the expressions for $P_{t+1}(x_{t+1}, y, z)$ and $B_{t+1}(x_{t+1})$ using Equations (15) and (12) respectively.

[19] Note that even if the worker receives a job after becoming unemployed next period, her value remains $B_{t+1}(x_{t+1})$ because the worker has zero bargaining power and only receives value offers equal to her reservation value.

$$
\begin{aligned}
w_t(x_t, y, z, \sigma_t) = {} & \sigma_t\, p_t(x_t, y, z) + (1-\sigma_t)\, b_t(x_t) \\
& - \beta(1-\xi)(1-\sigma_t)\, \mathbb{E}_t\Big[\mathbb{E}_\pi B_{t+1}(x_{t+1}) - B_{t+1}(x_t)\Big] \\
& - \beta(1-\xi)(1-\delta)\, \mathbb{E}_t \mathbb{E}_\pi\Big[s\, \lambda_{w,t+1} \mathbf{1}\{S_{t+1}(x_{t+1}, y, z) \geq 0\}\, \Omega_{t+1}(x_{t+1}, y, z, \sigma_{t+1})\Big], \quad (33)
\end{aligned}
$$

where the renegotiation term is

$$
\Omega_t(x_t, y, z, \sigma_t) = \int_{\mathcal{Y}} \left[\min\Big\{ S_t(x_t, y, z),\ \max\Big\{ S_t(x_t, y', z),\ \sigma_t S_t(x_t, y, z) \Big\} \Big\} \ -\ \sigma_t S_t(x_t, y, z) \right] \frac{v_t(y')}{V_t}\, dy'. \quad (34)
$$

Alternatively, the expression inside the large square brackets in $\Omega_t(x_t, y, z, \sigma_t)$ can be written as $\min\Big\{(1-\sigma_t)S_t(x_t, y, z),\ \max\Big\{0,\ S_t(x_t, y', z) - \sigma_t S_t(x_t, y, z)\Big\}\Big\}$, which can be interpreted as the change in a firm's share of the surplus when an outside offer arrives for the worker:

$$
\min\{S, \max\{S', \sigma_t S\}\} - \sigma_t S = \begin{cases} 0, & S' \leq \sigma_t S \quad \text{(offer below current value)} \\ S' - \sigma_t S, & \sigma_t S < S' \leq S \quad \text{(worker stays, renegotiates up)} \\ (1-\sigma_t)S, & S' > S \quad \text{(worker leaves, firm loses its full share)} \end{cases}
$$

where $S = S_t(x_t, y, z)$ and $S' = S_t(x_t, y', z)$. In Equation (33), the first term denotes the share of surplus that the firm pays to the worker, over and above the worker's flow value of unemployment $b_t(x_t)$. The second term denotes a worker's willingness to accept a lower wage, since she understands that employment leads to skill appreciation that raises her value of unemployment in the future. The third term says that the firm anticipates outside offers in the future that might lead to wage increases and/or job switches, and it compensates for this possibility by lowering the wages offered in the current period.

IV Parametrization

To understand the impact of an increase in soft skill demand on labor market outcomes, we will first fit the model to key aggregate moments for the US labor market between 1951 and 2011. In this section, we document how well our model fits the data, and the parameter estimates we obtain.

The model is solved at a weekly frequency, and simulated model moments are aggregated to the monthly or quarterly level to match the frequency at which the data moments are available.[20] We set the discount rate to an annual rate of 5%, and matching elasticity ω is fixed at 0.5. Worker skills x_t and firm types y are each approximated using a linearly spaced grid having 21 points on [0,1]. Firm productivities are obtained from a uniform distribution over

[20] The reason why we choose a weekly model frequency is to reduce time aggregation bias. For example, if we decided to simulate the model at a monthly or lower frequency, we would be unable to capture the labor market transitions properly for a worker who loses her job at the beginning of the month and obtains another job at the end of the month. At a monthly or lower frequency, this would be recorded as an EE transition although in reality it is an EU and UE transition. At a weekly frequency, we are more likely to identify the flows correctly.

[0,1]. Each period, new workers enter the labor force and draw their initial skill level from a Beta(2,12) distribution. The exit rate ξ corresponds to a worker remaining in the labor force for 40 years on average. The aggregate productivity shock z follows an AR(1) process, set up in the model using the discretized Rouwenhorst method, where mean$(z_t) = 1$.

We will target the means and standard deviations of labor market stocks and flows, such as the unemployment rate (U), and the transitions from unemployment to employment (UE), employment to unemployment (EU), and job-to-job (EE). Aggregate time series data on most of the above moments are obtained from BLS for our time period under consideration, except for job-to-job transitions, for which we rely on CPS data between 1994 and 2011 (after applying the corrections proposed by Fujita et al. (2024)).

For labor demand side moments, we utilize data on the standard deviation of vacancies created (V), and the standard deviation and autocorrelation of firm value added (VA). Data on vacancies is obtained from combining the BLS Help Wanted Index (HWI) for the period 1951-2006, and JOLTS for the period 2001-2011, where we project the HWI data on JOLTS data for the overlapping years to get a consistent time series. Time series data on firm value added is obtained through BLS, and we use Compustat data to calculate the mean and standard deviation of the cross-sectional dispersion in firm value added (Bloom et al., 2018).

We also construct moments for the mean and standard deviation of labor market tightness (V/U), and for the correlations between value added and the multiple labor market variables (U, V, UE, EU). Finally, we also utilize data on the correlation between V and U, between UE and EE, and between ΔVA and the cross-sectional dispersion in firm value added. The values for our data moments and the corresponding model moments are given in Table 4.

Table 4: Data Moments and Model Simulated Moments

Fitted moment	Data	Model	Fitted moment	Data	Model
$\mathbb{E}[U]$	0.058	0.060	sd$[U]$	0.191	0.175
$\mathbb{E}[UE]$	0.421	0.451	sd$[UE]$	0.127	0.131
$\mathbb{E}[EU]$	0.025	0.028	sd$[EU]$	0.100	0.069
$\mathbb{E}[EE]$	0.025	0.017	sd$[EE]$	0.095	0.105
$\mathbb{E}[V/U]$	0.634	0.091	sd$[V/U]$	0.381	0.355
$\mathbb{E}$[sd labor prod]	0.494	0.595	sd[sd labor prod]	0.039	0.025
sd$[V]$	0.206	0.185	corr$[V, VA]$	0.721	0.981
sd$[VA]$	0.033	0.034	corr$[UE, VA]$	0.878	0.951
autocorr$[VA]$	0.931	0.985	corr$[EU, VA]$	−0.717	−0.773
corr$[V, U]$	−0.846	−0.951	corr$[UE, EE]$	0.695	0.976
corr$[U, VA]$	−0.860	−0.951	corr[sd labor prod, ΔVA]	−0.366	−0.318
Annual wage growth	0.042	0.039			

Notes: UE, EU and EE are the monthly unemployment-to-employment, employment-to-unemployment and job-to-job transition rates; VA is aggregate value added, ΔVA its annual growth rate, and "sd labor prod" the cross-sectional standard deviation of log value added per job. Annual wage growth is the average change in log wages between the ends of consecutive years of experience (years 2 to 30), for workers employed at both dates; its data counterpart is 1.21/29 (Elsby and Shapiro, 2012). The model is solved and simulated at a weekly frequency on a $21 \times 21 \times 51$ grid over (x_t, y, z) for 700 years, the first 100 of which are discarded, and the simulated series are then aggregated to the quarterly frequency (annual for the productivity-dispersion moments) exactly as in the data.

Table 5: Parameter Estimates

		Estimate			Estimate
Matching	α	0.8105	Value added	a_1	0.0295
$\alpha L^{\omega} V^{1-\omega}$			$p(x_t, y, z) = z\big(a_1 + a_2 x_t + a_3 y$	a_2	−0.3000
Search efficiency	s	0.0840	$+ a_4 x_t^2 + a_5 y^2 + a_6 x_t y\big)$	a_3	0.8657
Vacancy costs	κ_0	0.2925		a_4	10.3074
$\frac{\kappa_0}{1+\kappa_1} v(y)^{1+\kappa_1}$	$1+\kappa_1$	1.0817		a_5	−3.2149
Exogenous separation	δ	0.0078		a_6	7.6516
Productivity shocks	σ_z	0.0472			
	ρ	0.9994	Skill accumulation	π	0.00035

Notes: This table shows the estimated parameter values from the simulated method of moments exercise. The discount rate is fixed at 5% per annum. The model is estimated at a weekly frequency. ω is fixed at 0.5. π is the weekly probability that an employed worker moves one rung up the skill ladder.

To match the model to data moments, we use simulated method of moments (SMM). The SMM objective function is nonsmooth and has many local minima, so we use a mix of global search (using differential evolution) and local refinement (using Nelder-Mead) to obtain the set of parameter estimates that minimize the objective function. We simulate the model for 700 years (which includes an initial burn-in period of 100 years to minimize the effect of initial conditions). Table 5 shows the parameter estimates we obtain from the simulation.

Even though none of the parameters have a clear one-to-one relationship with a moment, we can provide a rough description of which set of moments have a major influence on the identification of which parameters. The parameters for matching efficiency (α), relative search efficiency of employed workers (s), and exogenous separation rate (δ) govern movements in the labor market in the model, and they are identified primarily by the moments related to labor market flows ($\mathbb{E}[UE], \mathbb{E}[EE]$, and $\mathbb{E}[EU]$ respectively). The parameters related to the aggregate shock process (ρ, σ_z) are identified by the standard deviation and autocorrelation of output.

The standard deviation of vacancies and the correlation of vacancies with output help us identify the vacancy cost function parameters (κ_0, κ_1). Firm heterogeneity is identified by the cross-sectional variation in value added per job. The remaining standard deviation and correlation moments help us identify the production function parameters that govern complementarities between the firm and worker type.

We find an aggregate matching efficiency of 0.81 at the weekly frequency. Relative search efficiency of the employed workers is 0.084, which also accounts for the fact that not all employed workers search for jobs in all periods. The small, positive value of κ_1 indicates that the vacancy cost function is slightly convex. Jobs separate at an exogenous rate of 0.0078 every week. The aggregate shock process is very persistent at the weekly frequency, with $\rho = 0.999$.

The speed of on-the-job skill accumulation (denoted by π) is identified by the average growth in wages over 30 years of work experience. This is equal to 1.21 log points (Elsby and Shapiro, 2012), indicating a worker's wages grow by 4.2% per year of experience on average, and 3.35 times a worker's initial wages after 30 years of work experience. We estimate $\pi = 0.00035$ per week, which generates annual wage growth of 3.9% in the model.

The worker distribution is shown in Figure 7a. The skewness of the distribution indicates that low-skilled workers are more common in this labor market. The vacancy distribution (as shown in Figure 7b) is also right-skewed, but it peaks at higher firm types than the worker distribution peaks in skills. This is because the firm's vacancy creation decision depends not only on the skill levels of the workers, but also on their availability. This figure also shows that if the economy is in a productive state (high aggregate shock), more vacancies will be created since a higher surplus is generated in this state.

The production function parameters give us a key insight into the functioning of the model. a_1 is small, indicating that barely any output is produced by matches between the lowest firm and worker type. Keeping the firm type fixed, production is increasing for almost all worker types due to the large and positive value of a_4. Keeping the worker type fixed, output is non-monotonic in the firm type ($a_5 < 0$). For worker types $x_t < 0.73$, this implies a lower output produced if they match with firms of a relatively higher type.

Figure 8 shows the surplus generated by matches between different combinations of firms and worker types, calculated at the state when $z = z_{mid} = 1$. The white lines denote the boundary beyond which surplus becomes negative. The dashed red line gives the firm type y^* that maximizes the output produced for each worker type x_t. $a_5 < 0$ implies that y^* exists (even though it might be at the corner for some worker types), and $a_6 > 0$ implies that there is positive complementarity between the firm and worker types, leading to y^* being different for different worker types.

Figure 8 indicates how sorting takes place in this labor market. For low-type workers, matches with high-type firms are unviable. Intermediate worker types continue to switch employers until they reach as close to their maximum output generating firm type y^* as possible. For high-type workers, surplus is increasing in the firm type, so they are most likely to match with high-type firms.

V Counterfactual

In our counterfactual exercise, we analyze how much the unemployment rate of young workers rises relative to that of older workers in our model when we introduce a labor productivity shock similar in magnitude to the rise in soft skill demand during the period 2011-2024.

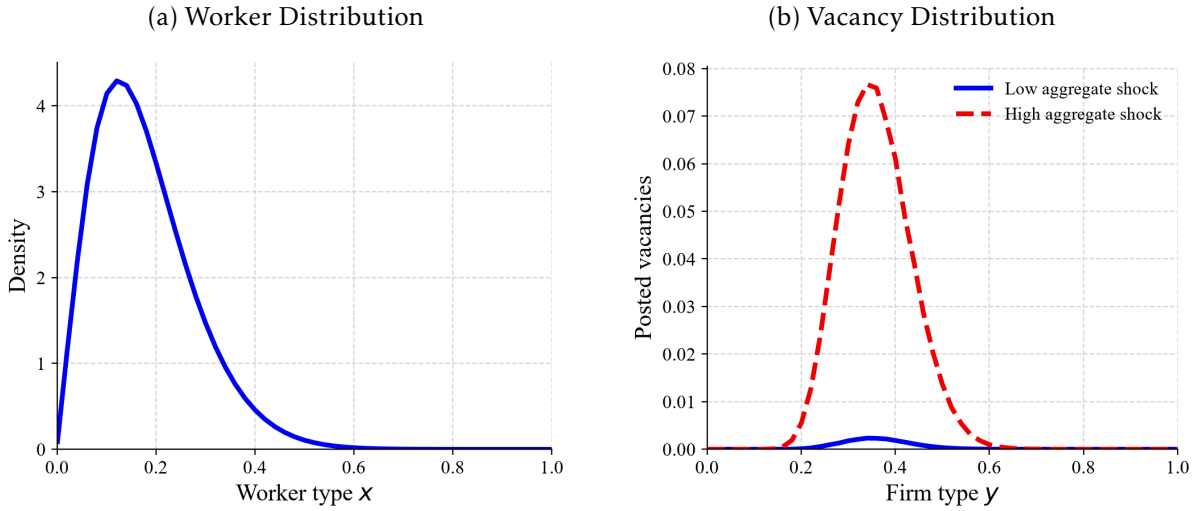
Given the estimated parameters, we simulate a labor market where we also keep track of the time each worker has spent in the labor force since entering it. In this simulated labor market, we define young workers as those who entered the labor market at most four years ago (consistent with college graduates between 22 and 25 years old), and older workers as those who entered more than four years ago. Since workers accumulate skills with work experience, young workers in this labor market have lower skill levels on average than older workers.

We model the rise in skill demand as a positive labor productivity shock. Consider the following modification of the production function in Equation 5:

$$p_t(x_t, y, z) = z_t\Big(a_1 + a_2(mx_t) + a_3 y + a_4(mx_t)^2 + a_5 y^2 + a_6(mx_t)y\Big),$$

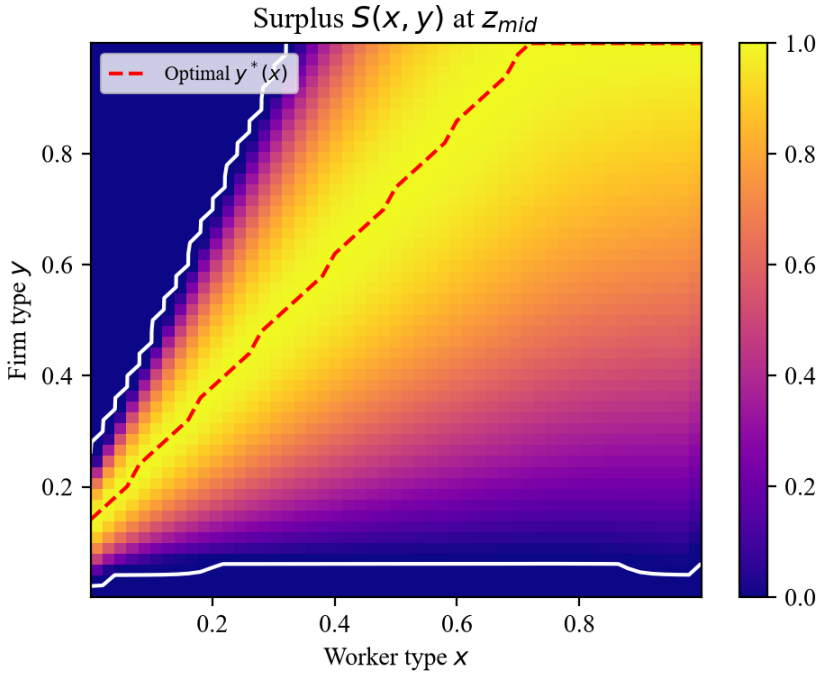
where m denotes the effective productivity of a worker's skills. In the baseline model, $m = m_{2011} = 1$. We set the increase in m equal to the cumulative increase in soft skill demand that

Figure 7: Worker Types and Vacancy Creation



Notes: Panel (a) plots the steady-state distribution of worker skills x_t generated by the estimated parameters of the model, when $z = z_{mid} = 1$. Panel (b) plots the distribution of vacancies created by firm types. The low (high) aggregate shock is equal to the 10th (90th) percentile in magnitude of the shock process, holding the distribution of workers at its steady state. All model figures are computed on a 51×51 grid over (x_t, y) at the parameter estimates in Table 5, with π rescaled so that the expected weekly skill gain is the same as on the 21×21 estimation grid.

Figure 8: Match Surplus



Notes: This figure portrays the surplus generated by all possible worker-firm matches when there is no aggregate shock in the labor market ($z = z_{mid} = 1$). The white lines in the top left and bottom right areas delineate the boundary where matches create zero surplus. The dashed red line denotes the firm type $y^*(x_t, 1)$ which maximizes the output produced by a worker of type x_t.

Table 6: Counterfactual Estimates

	2011	2019	2024
Data	1.91	2.33	2.57
Model	1.81	1.89	1.93

Notes: This table shows the ratio between the unemployment rates of young and older workers in different years. In the data, Young = 1 if age ≤ 25, and 0 if age $\in [26, 55]$. The data values are calculated from the CPS. In the model, young workers are those who entered the labor market at most four years ago, and older workers are those who entered more than four years ago. The 2011 model value is the unemployment rate ratio in the baseline model ($m = 1$). The 2019 (2024) model value is the ratio when the effective productivity of skills m rises by the increase in soft skill demand between 2011 and 2019 (between 2011 and 2024), i.e., $m = 1.10$ ($m = 1.16$). Model values are averages across 1,000 simulated paths of the aggregate shock.

we document in Section II. Soft skill demand increased by 10% between 2011 and 2019, and by 16% between 2011 and 2024, so we set $m_{2019} = 1.10$ and $m_{2024} = 1.16$ in the counterfactual. Under this specification, the productivity gain is larger for higher-skilled workers:

$$\begin{aligned} \frac{\partial p_t(x_t, y, z)}{\partial m} &= z_t(a_2 x_t + 2a_4 m x_t^2 + a_6 x_t y) \\ \Rightarrow \frac{\partial}{\partial x_t}\left(\frac{\partial p_t(x_t, y, z)}{\partial m}\right) &= z_t(a_2 + 4a_4 m x_t + a_6 y) \\ &> 0, \end{aligned}$$

where the inequality holds at all but the lowest skill level and firm type in our simulated labor market, given the values of a_2, a_4, and a_6 in Table 5.

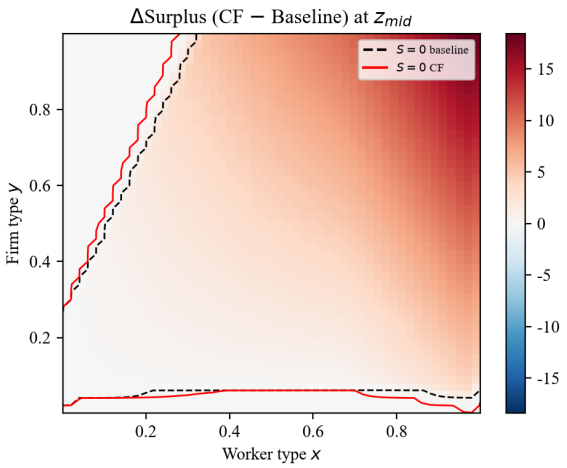
For each value of m, we hold all other parameters fixed at their estimated values, solve for the new surplus function, and simulate the stochastic equilibrium of the model. Specifically, we draw 1,000 independent paths of the aggregate productivity shock, each covering 700 years at a weekly frequency, and discard the first 100 years of each path.

Our object of interest is the ratio between the unemployment rate of young workers and the unemployment rate of older workers. We calculate this ratio both in the data and in the simulated labor market for the years 2011, 2019, and 2024, and report the values in Table 6. In the data, the ratio rises from 1.91 in 2011 to 2.33 in 2019 and 2.57 in 2024. In the model, it rises from 1.81 in 2011 to 1.89 in 2019 and 1.93 in 2024. The model (data) indicates an increase of 0.08 (0.42) in the ratio between 2011 and 2019, and an increase of 0.12 (0.66) between 2011 and 2024. Therefore, the rise in skill demand in our model generates approximately 19% of the observed increase in the relative unemployment rate of young workers during the pre-COVID period 2011-2019, and about 18% of the observed increase over the period 2011-2024. Measured in percentage changes instead of levels, the model accounts for about 20% and 19% of the observed increase over the two periods.

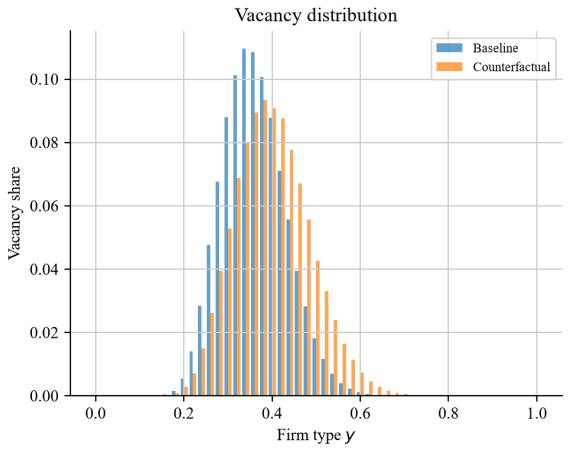
Why does an improvement in labor productivity lead to a relative rise in youth unemployment? Figure 9 illustrates the mechanism. The labor productivity shock makes all worker types more productive, and the set of viable matches (matches with a surplus ≥ 0) weakly expands for all low-skilled workers. Firms also create more vacancies as more surplus is generated by the same worker type in a match. However, since the productivity gain increases

Figure 9: Counterfactual: Change in Surplus and Vacancy Creation

(a) Change in Match Surplus

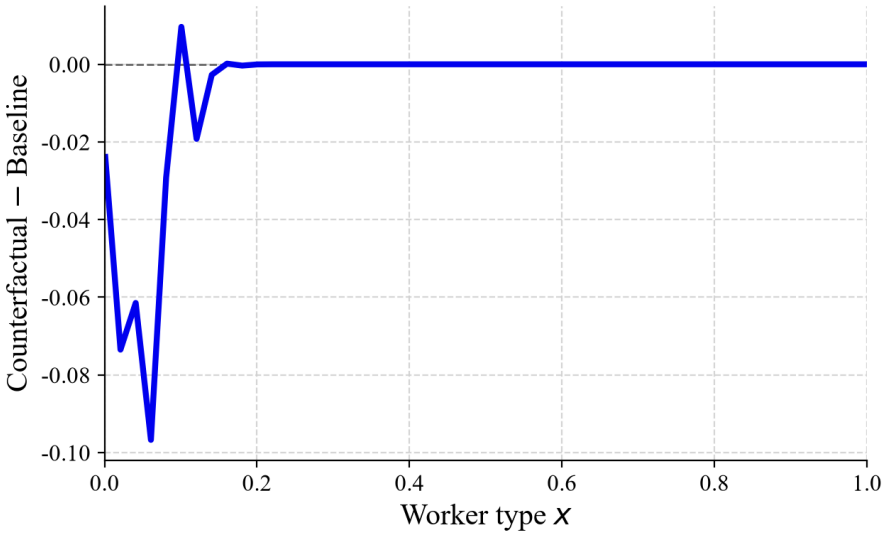


(b) Change in the Vacancy Distribution



Notes: Panel (a) shows the change in match surplus and in the set of viable matches after an increase in labor productivity ($m = 1.16$), when $z = z_{mid} = 1$. The dashed black lines denote the boundaries of the original set of viable matches in the baseline model. The solid red lines show the new boundaries after labor productivity increases. Panel (b) shows the vacancy distributions before and after the increase in labor productivity, each at its own steady state when $z = z_{mid}$. Both panels use the 51×51 grid described in Figure 7a.

Figure 10: Fewer Viable Vacancies For Low-Skilled Workers



Notes: This figure shows the change in the share of viable vacancies available for each worker type $x_t \in [0,1]$ in the counterfactual setting when labor productivity has increased ($m = 1.16$), when $z = z_{mid} = 1$. A vacancy from a firm type y is viable for a worker if the worker-firm match generates a positive amount of surplus. The share is weighted by each firm type's share of posted vacancies.

with skill, high-skilled workers become relatively more productive, and the complementarity between firm types and worker skills means that high-type firms experience a greater rise in the value of filling a vacancy than low-type firms. Therefore, there are relatively more vacancies from high-type firms after the labor productivity shock hits, and the vacancy distribution shifts to the right. Since young, low-skilled workers do not generate a positive match surplus with high-type firms, a smaller share of the vacancies they encounter is viable (Figure 10). This generates a relative increase in the unemployment rate of young workers.

VI Conclusion

This paper set out to understand why young college graduates have fared increasingly poorly in the US labor market. We document three empirical facts. First, since 2011, the unemployment rate of young college graduates has been rising relative to that of the average US worker. Second, using near-universe data on online job postings, we show that employer demand for soft (noncognitive) skills has risen since 2011. Third, the gap between the wage of young and older workers is larger in jobs that are more soft-skill intensive. Combining these facts, we argue that as jobs become more soft-skill intensive over time, young workers face increasingly worse labor market outcomes, observed in the data as an increase in their unemployment rate.

We discuss that young workers fare worse in soft skill intensive jobs because soft skills are accumulated on the job, so young workers have lower levels of these skills when they enter the labor market. We build a general equilibrium, search-and-matching model, where we show that an increase in soft skill demand can account for one-fifth of the increase in the unemployment rate of young workers relative to older workers.

Thus far, our focus has been on the unemployment rate. The rise in youth unemployment is analogous to an entry barrier to the labor market. It is possible that once young college graduates receive a job, they are able to acquire the relevant soft skills and do well in the labor market. However, similar to the related literature that finds long-run scarring effects on workers entering the labor market during a recession, we may find long-run negative effects of an increase in soft skill demand as well. Therefore, in future work, we also want to analyze the effect of soft skill demand on lifetime earnings.

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Appendix

A Empirical Analysis

A1 Data and Measurement

This appendix describes sample construction and measurement and reports the descriptive tables and figures for Section I. It covers the CPS and MORG samples, Lightcast posting characteristics, and the comparison of advertised skill counts with O*NET occupational ratings.

A1.1 CPS and MORG Samples

The descriptive sample in this appendix covers 2011–2024 and compares workers aged 18–25 with those aged 26–55. College-educated workers hold a bachelor's degree or higher; all others are classified as noncollege. Age identifies young degree holders rather than graduation dates or actual work experience. These descriptive statistics do not describe the exact sample used in the wage regressions: those regressions cover ages 16–54, use real hourly earnings, and apply composition-adjusted final CPS weights. Table A7 reports the sample and weights for those estimates. The separate labor market flow analysis uses ages 21–29 and 30–55.

Table A1 reports descriptive statistics for 2011–2024, the common period covered by the CPS and Lightcast inputs used here. Panel A describes the basic monthly CPS population, including part-time workers, the self-employed, unemployed people, and people outside the labor force. Employment includes wage employment and self-employment; the unemployment rate divides the unemployed by the labor force. We use final CPS survey weights and exclude observations with unknown employment status from these rate denominators. The cleaned CPS file harmonizes demographic and employment categories and excludes military observations.

Panel B describes the full-time MORG wage sample with Lightcast skill requirements assigned by occupation and month. It retains wage and salary workers with at least 35 hours at their main job, positive hourly earnings, and positive ORG earnings weights. The hours measure uses usual main-job hours where available, with substitutions otherwise. Hourly earnings are weekly earnings divided by hours, with reported hourly pay used when that ratio is unavailable. We retain allocated and top-coded earnings. Both hourly and weekly earnings in the descriptive tables are expressed in 2024 dollars using the personal consumption expenditures price index.

This wage sample contains 1,288,565 observations. These are person-months: the CPS can interview the same person in several months. Wage statistics use ORG earnings weights, while the population statistics use final CPS survey weights. The cleaned age-restricted sample contains 9,460,549 person-months, of which 9,193,824 have observed education. The outgoing-rotation subsample contains 2,335,723 observations; requiring wage employment, at least 35 main-job hours, positive hourly earnings, and positive earnings weights leaves 1,329,640 before requiring available Lightcast requirements for the respondent's occupation and month. This restriction retains 96–98% of eligible earnings weight across the four age–education groups. Allocated earnings account for 32–37% of this wage sample's earnings

weight. Weekly earnings are flagged as top-coded for 2% of young college workers, 11% of older college workers, less than 1% of young noncollege workers, and 1% of older noncollege workers. These observations are retained; sample coverage alone does not establish that the wage results are insensitive to earnings allocation or top coding.

Assigning Occupational Requirements in the Wage Analysis For the wage analysis and the assigned-requirement statistics in Panel B, we map Lightcast and CPS occupation codes to common occupation groups. Within each group and month, we sum postings and skill mentions before taking their ratio. Occupations therefore contribute in proportion to their posting counts. We then merge these occupation–month averages into the CPS wage sample. Each respondent receives the average advertised requirements for her occupation and month, rather than the requirements of a particular vacancy. The posting-only exhibits below use occupation codes to group Lightcast advertisements; their construction does not require linking postings to CPS observations.

Table A1: Workers, Earnings, and Assigned Skill Requirements

	College 18–25	College 26–55	Noncollege 18–25	Noncollege 26–55
A. Basic monthly CPS				
Observations	261,032	2,786,779	1,528,124	4,617,889
Mean age	23.6	40.0	21.2	40.6
Women (%)	57.5	54.2	48.4	48.7
Employment/population (%)	77.9	86.9	57.3	73.6
Full-time wage employment/population (%)	61.0	71.8	31.5	56.9
Labor-force participation (%)	83.0	89.3	64.7	78.5
Unemployment rate (%)	6.2	2.7	11.4	6.1
B. Full-time MORG with occupation–month skill requirements				
Observations	39,294	495,521	116,755	636,995
Share with skill requirements (%)	96.3	97.9	96.5	96.3
Mean hourly earnings (2024 dollars)	28.06	43.35	17.65	25.61
Median hourly earnings (2024 dollars)	24.76	38.22	15.97	22.00
Mean weekly earnings (2024 dollars)	1,161	1,856	724	1,080
Mean weekly hours	41.4	42.9	40.8	42.0
Assigned soft requirements: mean	3.04	3.16	2.22	2.46
Assigned soft requirements: SD	1.24	1.35	1.19	1.35
Assigned technical requirements: mean	10.56	10.89	7.24	8.27
Assigned technical requirements: SD	4.32	4.38	2.98	3.52

Notes: Sample period: 2011–2024. College denotes a bachelor's degree or higher. Observations are person-months, not unique people. Panel A uses final CPS survey weights and retains part-time workers, the self-employed, unemployed people and people outside the labor force. Employment includes wage and self-employment. Labor-market rates exclude unknown employment status; unemployment is divided by the labor force. Panel B uses ORG earnings weights and requires wage employment, at least 35 main-job hours, positive hourly earnings and available Lightcast requirements for the occupation and month. Earnings are PCE-deflated to 2024 dollars. Allocated and top-coded earnings are retained. Assigned requirements are average skills per posting in a worker's occupation-month, not measured worker skills. SDs describe dispersion across workers; the two panels have different samples and weights.

A1.2 Lightcast Postings and Skill Requirements

Posting Sample We measure employers' demand for skills using online job postings from Lightcast, formerly Burning Glass Technologies (Lightcast, 2024). Lightcast is a private business analytics company that collects postings from more than 160,000 sources worldwide each day. Its machine-learning algorithms extract information about the employer, location, occupation, industry, minimum experience, and posting date. For our purposes, the central feature is the list of skills requested in each posting.

The Lightcast data span January 1, 2011 through December 31, 2024. We exclude internships and postings with missing dates, occupations, industries, job titles, firm names, or skill lists. The sample for the descriptive exhibits contains 307,456,240 retained postings, 769 occupation codes, and 124,480 occupation-month cells. The sample also excludes military occupations, Craigslist-only postings, and selected unusable employer names. A posting need not represent one position or a realized hire. Our aggregated data do not contain the identifiers needed to independently check deduplication or report attrition at each sample restriction.

Skill Classification The April 2025 Lightcast taxonomy contains 34,350 distinct skill labels. The confirmed soft-skill set contains 397 labels, selected using the category and named-label rules in Appendix A1.2 below.

This is a taxonomy-based count measure. It differs from classifying the text of an advertisement, which is described below. The technical category is a residual and includes more than narrowly technical expertise.

Skill Counts and Occupational Aggregation The skill-count measures use the following rules from the April 2025 Lightcast taxonomy. Soft skills include the category Physical and Inherent Abilities except its Physical Abilities subcategory; Communication within Media and Communications; and People Management, Business Leadership, and Business Communications within Business. These rules include common and specialized skills and exclude certifications. Within Business Management, only common skills enter the soft category. We also include the individual labels Public Relations, Crisis Management, Sales, Retail Sales, Negotiation, Operations, and Consulting. All remaining labels enter the technical count, including certifications. The categories partition the distinct labels in each posting.

We measure skill demand as the average number of skills mentioned per posting, calculated separately for soft skills, technical skills, and all skills. Each distinct skill label is counted once within a posting.[21] For posting j, let S_j and T_j denote the soft- and technical-skill counts. For occupation o in month t, we construct

$$S_{ot} = \frac{1}{N_{ot}} \sum_{j \in (o,t)} S_j, \qquad T_{ot} = \frac{1}{N_{ot}} \sum_{j \in (o,t)} T_j, \tag{35}$$

[21] The measure gives equal weight to every skill mentioned in a posting. In Section A3, we examine whether O*NET can help account for differences in skill importance. The appendix shows that O*NET captures relative skill requirements across occupations, but does not reproduce the aggregate growth in requirements observed in Lightcast. It also discusses using AI with O*NET ratings to classify the soft- and technical-skill intensity of Lightcast postings.

where N_{ot} is the number of retained postings. The average total skill count is $S_{ot} + T_{ot}$. A posting with an observed skill list but no soft skills contributes zero to S_{ot}; postings without a skill list are excluded.

Table A2 reports the distributions in the 307,456,240-posting sample, using the taxonomy-based classification described above. Each posting receives equal weight within a period. Postings request an average of 3 soft skills and 10.2 technical skills, and 75% mention at least one soft skill. The period comparison separates 2011–2015 from 2016–2024 and retains every available year. The posting counts end in December 2024.

Skill counts measure the breadth of advertised requirements. They do not measure required proficiency or workers' own skills. Differences in the granularity of skill labels also limit comparisons between soft and technical counts: several closely related technical labels can appear in the same posting. The larger technical count therefore need not imply greater economic importance.

Coverage and Composition Table A3 reports the availability of education and experience requirements. Figure A1 shows annual posting counts. Figure 5 in the main text reports skill intensity in the separate classified-advertisement sample, using rolling twelve-month windows and a fixed three-digit occupational mix. Appendix A2 describes the text sample and scoring rules.

The occupational mix is not constant over this period. Under the fixed 2011 classification, the share of postings in soft-skill-intensive occupations falls from 74% to 58%, and the share in technical-skill-intensive occupations falls from 74% to 63%. To examine changes beyond these two groups, let p_{ot} denote occupation o's share of all retained postings in year t. The distance $D_t = \frac{1}{2}\sum_o |p_{ot} - p_{o,2011}|$ equals zero when the occupational distribution is unchanged. In 2024 it is 0.241 at the detailed occupation level and 0.177 after aggregation to two-digit SOC groups. These changes make it important to account for occupational composition when comparing advertised requirements over time.

Table A2: Job Postings and Advertised Skill Requirements

	2011–2015	2016–2024	2011–2024
Postings (millions)	62.56	244.90	307.46
Occupations	767	767	769
Occupation-month cells	43,740	80,740	124,480
Soft skills			
Mean	2.84	3.04	3.00
Standard deviation	3.02	3.13	3.11
Median	2.00	2.00	2.00
90th percentile	7.00	7.00	7.00
Postings mentioning any (%)	72.88	75.36	74.86
Mean, conditional on mentioning any	3.90	4.04	4.01
Technical skills			
Mean	9.84	10.29	10.20
Standard deviation	9.17	8.75	8.84
Median	8.00	8.00	8.00
90th percentile	21.00	22.00	22.00
Postings mentioning any (%)	98.86	99.07	99.02
Mean, conditional on mentioning any	9.95	10.39	10.30

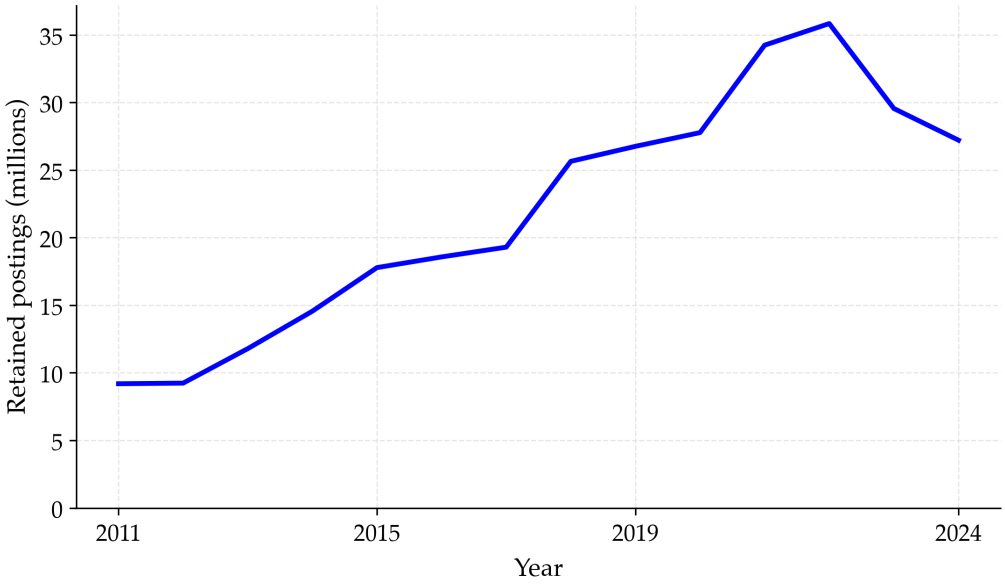
Notes: An observation is a retained posting. Each posting receives equal weight; repeated skill labels within a posting are counted once. The source excludes postings without an observed skill list, so zero means no skill in that category, not missing skills. Technical skills are the non-soft residual under the taxonomy-based classification. Percentiles use the inverse empirical distribution; standard deviations describe the posting distribution. Periods pool postings rather than weighting years equally. Counts need not represent unique positions or hires. Source: Lightcast, 2011–2024.

Table A3: Coverage of Education and Experience Requirements

	2011–2015	2016–2024	2011–2024
Postings (millions)	62.55	244.88	307.43
Missing education requirement (%)	44.00	45.49	45.19
Missing experience requirement (%)	51.41	53.08	52.74
Bachelor's or higher required (%)	30.99	25.48	26.60
Listed as full-time (%)	88.48	84.88	85.61

Notes: This table uses the separate education/experience-crossed source, containing 307,433,363 postings. It has 22,877 fewer postings than the main skill-count sample because the two samples are aggregated using different merges. Each percentage uses this table's own posting count. Missing requirements are not coded as zero; the college-required share uses all retained postings, including those with missing education. Full-time is Lightcast's category (more than 32 hours), which differs from the CPS threshold of at least 35 hours.

Figure A1: Number of Job Advertisements

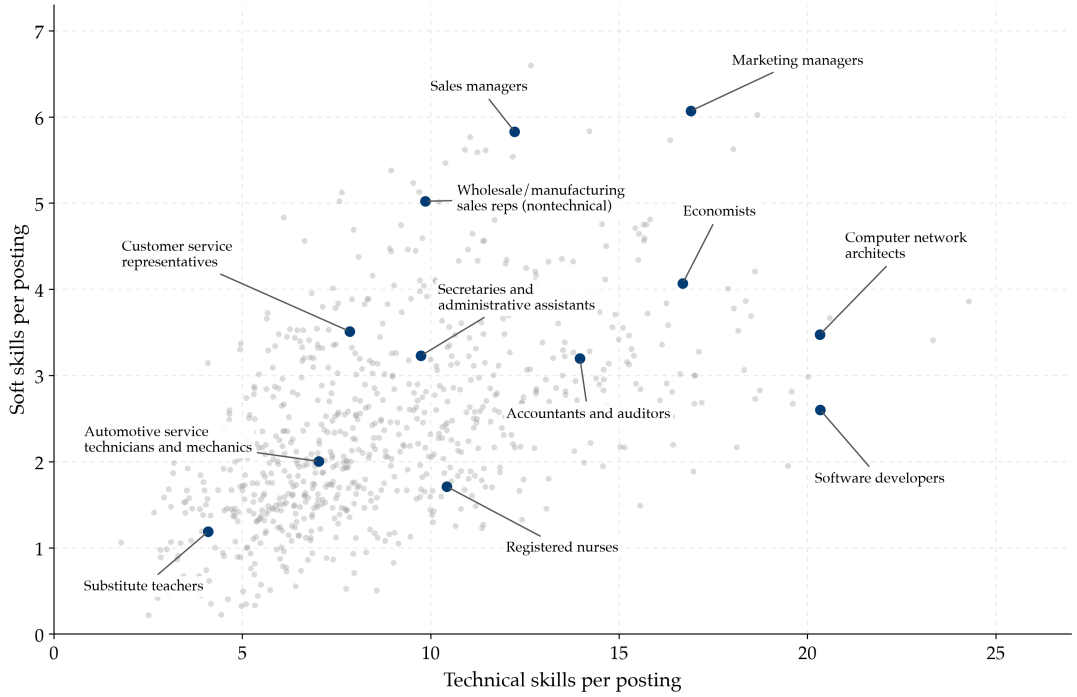


Notes: Annual counts of retained advertisements in the main Lightcast skill-count sample, 2011–2024. The sample contains 307,456,240 postings. These counts describe the full sample, rather than the subsample classified from ad text.

A1.3 Occupational Skill Requirements: All Occupations

Figure A2 extends the college-entry occupation map in Figure 1 to all occupations in the Lightcast skill-count sample.

Figure A2: Soft and Technical Requirements across All Occupations



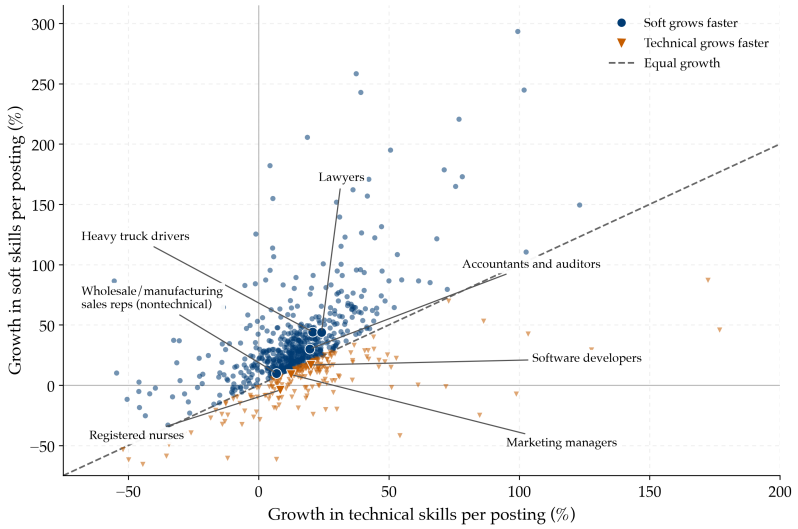
Notes: Each point represents one of the six-digit SOC occupations in Lightcast. The horizontal and vertical axes show the average numbers of technical and soft skills per posting, respectively, pooling 307,456,240 postings from 2011–2024. Skill mentions and postings are summed within occupation before calculating ratios, giving each posting equal weight. Soft skills follow the taxonomy rules in Appendix A1.2; technical skills comprise all remaining labels, including certifications. Blue points mark twelve illustrative occupations.

A1.4 Occupational Skill Requirements by Period

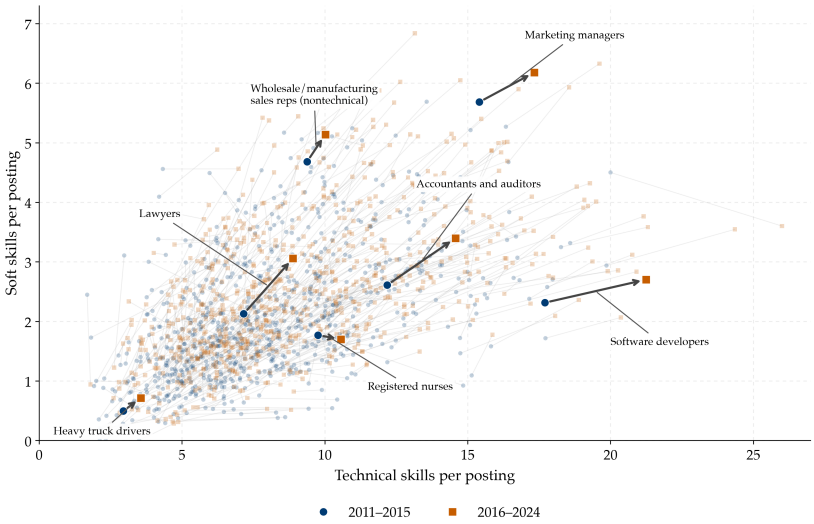
Figure A3 compares soft and technical requirements in 2011–2015 and 2016–2024. Panel A compares their growth within six-digit SOC occupations. Panel B shows the changes in levels for the same occupations.

Figure A3: Soft and Technical Requirements across Occupations by Period

Panel A: Growth within six-digit SOC occupations



Panel B: Levels across six-digit SOC occupations



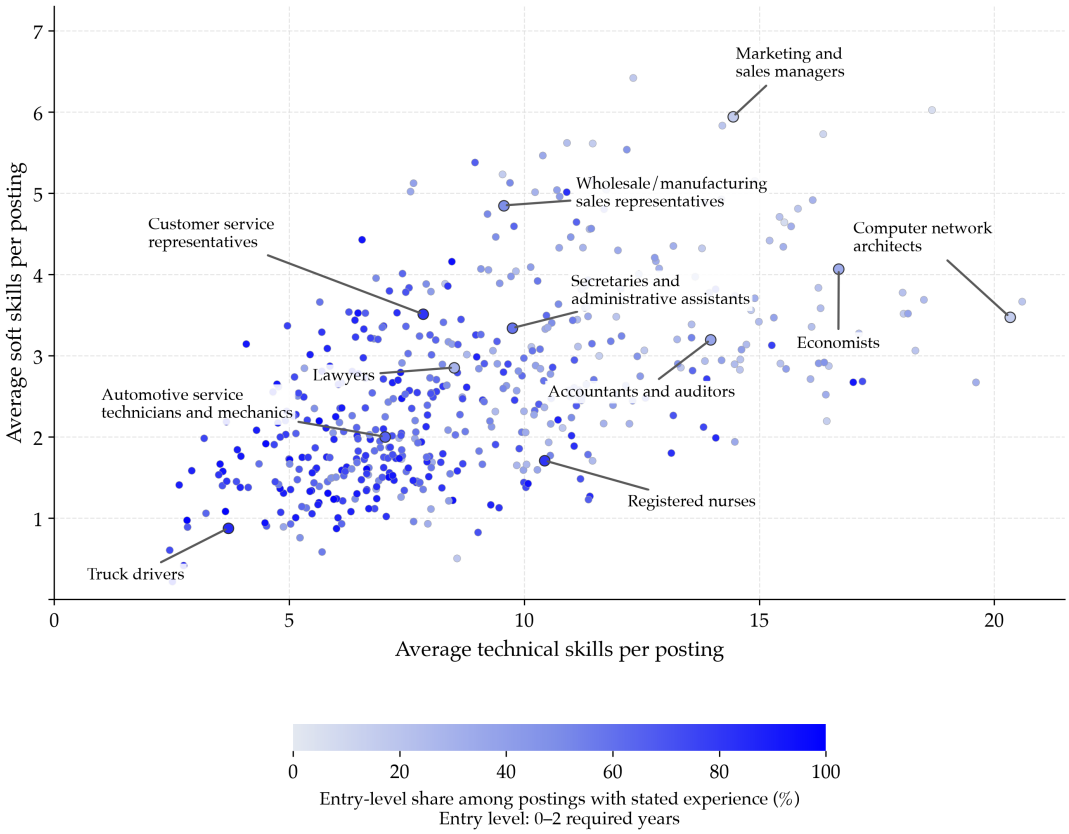
Notes: The comparison uses all 765 six-digit SOC occupations observed in both periods in the Lightcast skill-count sample. Four codes observed in only one period cannot be compared. Growth is the percentage change in mean skills per posting between the two periods, not an annualized rate. The scatter includes 762 occupations; blue circles above the dashed equality line indicate faster growth in soft skills. Three occupations with zero initial soft-skill mentions are omitted from Panel A because their percentage growth is undefined. Panel B includes all 765 occupations. Lines connect 2011–2015 (blue) to 2016–2024 (orange); arrows and darker markers highlight seven illustrative occupations, also labeled in Panel A. In both panels, means are ratios of total skill mentions to total postings within each occupation and period. Soft skills follow Appendix A1.2; technical skills comprise the remaining labels, including certifications.

A1.5 Occupational Skill Requirements and Entry-Level Postings

Figure A4 adds each occupation's share of entry-level postings to the all-occupation skill map in Figure A2. Among postings with stated experience requirements, the entry-level share is 81% for customer service representatives and 16% for marketing and sales managers. Both occupations have relatively high soft-skill counts, but their advertised experience requirements

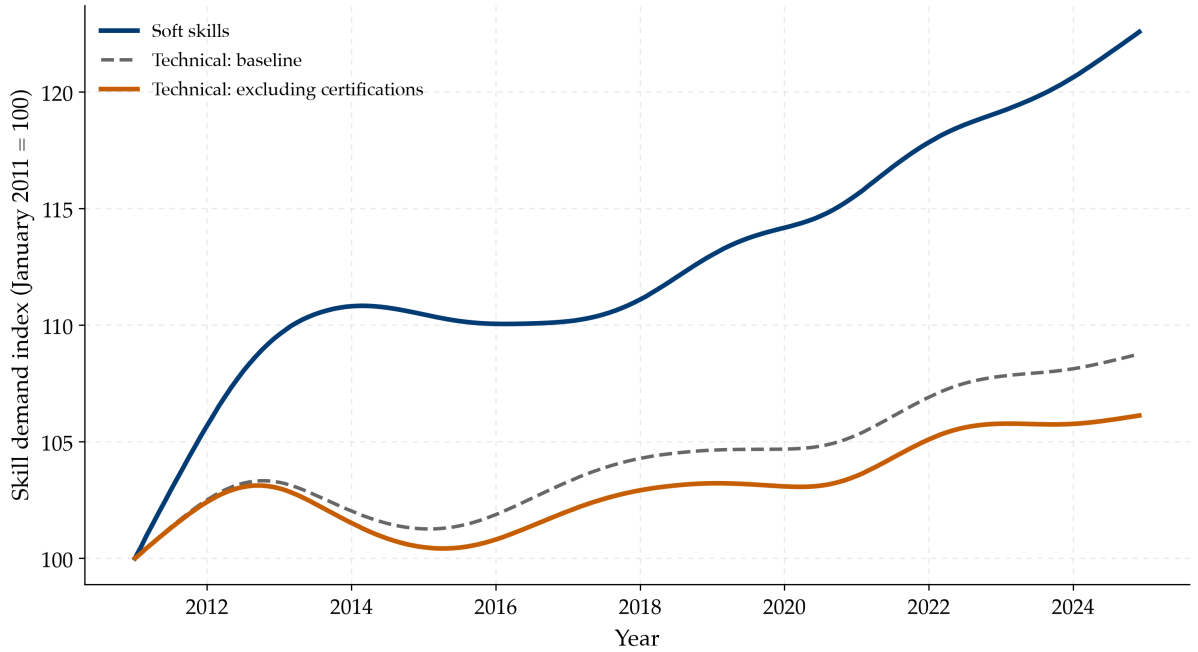
differ sharply.

Figure A4: Occupational Skill Requirements and Entry-Level Posting Shares



Notes: Each point represents one of 494 occupation groups formed by mapping Lightcast SOC codes through the occupation crosswalk. The sample pools 297,337,527 postings from 2011–2024 across all education requirements and employment types. The axes show average technical and soft skills per posting. Soft skills follow Appendix A1.2; technical skills comprise the remaining labels, including certifications. Darker blue indicates a larger entry-level share: postings requiring a minimum of 0–2 years of experience divided by all postings with stated experience in that occupation. Posting counts are summed across years before calculating the share. Experience is stated in 141,307,614 postings; the 52% with unspecified experience are excluded from the color calculation, while the skill averages retain postings with unspecified experience. Required experience describes the advertisement, not the worker ultimately hired. Outlined points mark the labeled occupations.

Figure A5: Skill Demand Trends with and without Certifications



Notes: Monthly means divide total skill mentions by total postings in the same Lightcast sample as Figure 3. The blue curve shows soft requirements; the gray dashed curve shows baseline technical requirements; the orange curve excludes certifications from technical counts. All three series use the baseline HP smoothing parameter of 14,400 and are normalized to 100 at their respective January 2011 trend values. No postings are removed when certifications are excluded. These aggregate series allow the occupational composition of postings to change over time.

A2 LLM Classification of Job Advertisements

We use Claude Sonnet 4.5 to measure the intensity of soft and technical requirements in individual Lightcast advertisements. The model reads the advertisement and assigns an independent low or high rating to each dimension. The exercise measures how skills enter the advertised role, complementing the skill-label counts used elsewhere in the paper. Figure 5 reports our preferred series, which holds three-digit occupational composition fixed and pools advertisements over rolling twelve-month windows. Panel A reports the two marginal high-rating indices, and Panel B reports the shares of all four joint classifications in levels.

Sample. The classified sample contains 100,128 advertisements dated January 2012 through December 2025, with 596 advertisements in each month and 7,152 in each year. Both ratings are available for 100,048 advertisements; 80 observations missing both ratings are excluded. Posting identifiers are unique. We use the supplied sampling weights in all reported shares. This text sample is separate from the full posting-count sample, which covers 2011–2024.

Joint distribution. Table A4 reports the pooled shares of all four soft–technical combinations in the full valid classified sample. Figure 5, Panel B, shows their time paths in levels, using the same fixed three-digit occupational composition and rolling twelve-month windows as Panel A.

Table A4: Joint Skill Classification in the Full Advertisement Sample

Combination	Advertisements	Weighted share (%)
H–H	33,597	34
H–L	23,029	23
L–H	17,012	17
L–L	26,410	27
Total	100,048	100

Notes: The first letter denotes soft skills and the second technical skills; H means high and L low. Shares use the supplied sampling weights and pool all advertisements with both ratings over 2012–2025, across all valid occupations. The 80 advertisements missing both ratings are excluded. No occupation or time standardization is applied. Percentages are rounded independently and may not sum to 100.

Classification prompt. The prompt asks the model to consider the full advertisement, including the title, seniority, experience, education, certifications, duties, and responsibilities. It defines the two ratings as follows:

> **Low:** "The ad contains little or no demand for the skill type, mentions it only in generic boilerplate, or requires only basic use that is not important to the role."
> **High:** "The skill type is important for the main duties, repeatedly emphasized, or explicitly required at an advanced level."

Soft skills include interpersonal and workplace capabilities: communication, teamwork, leadership, people management, client interaction, negotiation, stakeholder coordination, mentoring, planning, critical thinking, judgment, and decision making. Leading a team, advising clients, resolving conflict, or coordinating stakeholders provides substantive evidence of these requirements. Generic phrases such as "good communication skills" or "team player" receive a low rating.

Technical requirements concern occupation-specific knowledge applied through specialized methods, software, tools, equipment, systems, or procedures. The prompt covers programming, quantitative analysis, engineering, clinical and scientific methods, law, accounting and finance, auditing, and skilled trades. Examples of substantive use include designing, diagnosing, coding, analyzing, troubleshooting, and operating specialized equipment. Ordinary software use, data entry, and simple standardized procedures receive low ratings.

The prompt explicitly instructs the model to score the two dimensions independently. Evidence of technical requirements cannot lower the soft-skill rating, or vice versa. An advertisement can therefore be high on both dimensions, low on both, or high on only one; the high-intensity shares need not sum to one. Benefits, generic company-culture language, application instructions, equal-opportunity statements, and legal disclaimers are ignored. The model is instructed not to infer the project's desired result or a time trend. These are binary assessments of advertised requirements, rather than cardinal measures of worker proficiency.

O*NET context and implementation. The classifier supplies O*NET skill-importance and skill-level ratings matched to the advertisement's occupation and posting date. Reference rows are selected when the posting date falls within their validity interval. Importance ranges from 1 to 5 and level from 0 to 7. The prompt explains that these ratings describe the occupation

generally and may help interpret ambiguous duties; occupation or O*NET information alone cannot justify a high rating. The advertisement itself must supply the evidence.

We use Claude Sonnet 4.5 with temperature set to zero. Advertisements are submitted in batches, and the prompt requests one structured response per advertisement, retaining its identifier and separate soft and technical scores (1 for low and 2 for high). It also requests confidence assessments for each dimension and low confidence for vague, short, or inconsistent advertisements. We do not use these confidence assessments, and they do not determine the analysis sample.

Fixed occupational composition. Let H_i^k indicate a high rating for skill type k, and let w_i be the advertisement's sampling weight. Define the joint indicator as $H_i^{HH} = H_i^{\text{soft}} H_i^{\text{technical}}$. The other joint indicators are $H_i^{HL} = H_i^{\text{soft}}(1 - H_i^{\text{technical}})$, $H_i^{LH} = (1 - H_i^{\text{soft}})H_i^{\text{technical}}$, and $H_i^{LL} = (1 - H_i^{\text{soft}})(1 - H_i^{\text{technical}})$. The four cells of the classification are H–H, H–L, L–H, and L–L, with soft status listed first. Their shares sum to one; the soft-high share equals H–H plus H–L, and the technical-high share equals H–H plus L–H. The H–H denominator is all valid advertisements in the common sample, including those rated low in either dimension. We apply the same weighting and occupation adjustment to all cells. For each three-digit SOC group g and twelve-month window ending in month t, we calculate

$$p_{gt}^k = \frac{\sum_{i \in (g,[t-11,t])} w_i H_i^k}{\sum_{i \in (g,[t-11,t])} w_i}.$$

We retain the 85 groups represented in every twelve-month window. This common sample contains 99,603 advertisements and accounts for nearly 100% of the sampling weight among advertisements with valid ratings. Let s_{gy} be group g's weighted share of the common sample in calendar year y. The fixed reference share is $\bar{s}_g = \frac{1}{14} \sum_{y=2012}^{2025} s_{gy}$, giving each year equal weight. The aggregate share and plotted index are

$$P_t^k = \sum_g \bar{s}_g p_{gt}^k, \qquad I_t^k = 100 \frac{P_t^k}{P_{\text{Dec 2012}}^k}.$$

Thus, December 2012 represents the January–December 2012 sample, and each subsequent point advances the window by one month. We pool weighted counts within each window before calculating group shares; we do not average monthly percentages or apply an additional time-series filter. Holding the group mix fixed removes changes in the distribution across these broad occupations. It does not hold detailed jobs or employers fixed. The same formulas apply to $k \in \{HH, HL, LH, LL\}$. Figure 5, Panel B, plots the four P_t^k shares in levels. Panel A plots the marginal soft and technical indices, each using its own calendar-year 2012 share as the denominator.

Uncertainty and interpretation. The shaded bands are approximate pointwise 95% confidence intervals. Panel A's index bands are calculated by linearizing the log index. Panel B's share-level bands equal the adjusted share plus or minus 1.96 times its standard error. Both panels include uncertainty in the estimated reference occupation shares. Panel A additionally includes uncertainty in the 2012 baseline and retains covariance from advertisements shared

by overlapping windows. Posting months are treated as variance strata, with advertisements assumed independent within month. Our data do not identify the original sampling strata or employer clusters, so these are approximate sampling intervals. They exclude classification error and are not simultaneous bands for the entire time path. Some occupation–window cells are sparse.

The estimates describe changes in advertised requirements. They do not measure workers' skills or identify the effect of those requirements on employment. The H–H series measures joint advertised requirements; identifying complementarity in production would require additional evidence. No independent human-rating validation is reported for this sample. O*NET agreement alone would not provide such validation because O*NET information enters the prompt.

A3 O*NET and Lightcast

Our measure of an occupation's skill requirements (or skill demand) is calculated using Lightcast data on job postings. As we described in Section I, Lightcast is able to extract the skills that each job posting asks for. We classify these skills as soft or technical skills, and we define soft skill demand at the occupation × month level as the average number of soft skills per job posting within an occupation × month group (and similarly for technical skill demand). A drawback of using skills data from job postings is that we cannot observe the importance of each skill in a particular occupation. Instead, in our skill demand measure, all skills demanded in an occupation are weighted equally.

An alternative source of information on occupation skill requirements is the Occupational Information Network (O*NET, 2026). O*NET data consists of 35 skills and approximately 900 occupations, and it assigns an importance value between 1-5 and a level between 0-7 for each skill to each occupation. Importance signifies whether a skill is required in an occupation, and the level signifies the expertise in a skill that is needed in the occupation. For example, O*NET explains that the speaking skill is important for both lawyers and paralegals. However, lawyers are required to have a higher level in this skill than paralegals.[22] In this section, we explore O*NET skills data to understand its suitability for our analysis.

O*NET updates its data on occupation skill requirements each year to capture any changes in the tasks involved in an occupation over time. We create binned scatterplots to compare how skill requirements have changed in occupations between 2011 (Version 16.0) and 2024 (Version 30.2). In Figure A6 and Figure A7, we see that for both the importance and level plots, the points lie close to the 45 degree line for both soft skills and technical skills, respectively. This would suggest that the O*NET occupational skill requirements remain relatively unchanged over the time period of our analysis.

Does this mean that skill demand is stable over our time period, and that our Lightcast measure of skill demand is incorrect? We argue that it does not. Figure A8 shows binned scatterplots where we use our skill demand measure from Lightcast (average skills per posting in an occupation) instead of the O*NET skills requirement data. We observe that there is a near uniform increase in skill demand (both soft and technical) across occupations between 2011

[22] See the data dictionary for O*NET Version 30.2 and O*NET's definition of soft skills.

and 2024, irrespective of the skill demand in an occupation in 2011.

O*NET data may be unable to capture this near-uniform increase in skill demand because of two potential reasons. Both reasons relate to the scales used to measure the importance and level of skill requirements in O*NET. First, given that the scales are fixed (1-5 for importance and 0-7 for level), an increasing trend in skill demand would eventually lead to all skill requirements reaching the upper bound of these scales if O*NET attempted to include this information in its measures. Therefore, we believe that although O*NET data is useful for comparing skill requirements across occupations at a particular time, it is not suitable for understanding the trends in these skill requirements over time. In short, O*NET should be used for studying relative skill demand across occupations, not absolute skill demand.

Figure A6: O*NET Soft Skill Requirements (2011-2024)

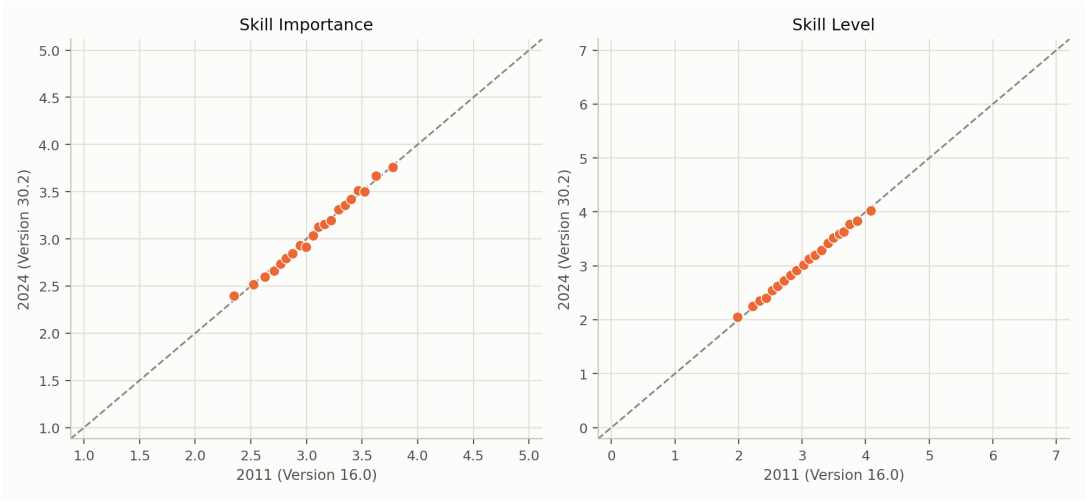
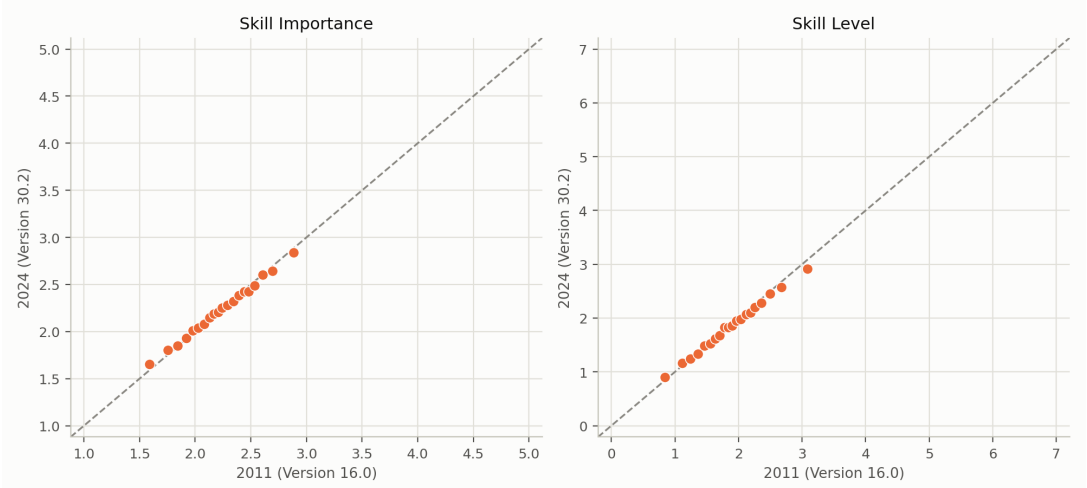


Figure A7: O*NET Technical Skill Requirements (2011-2024)

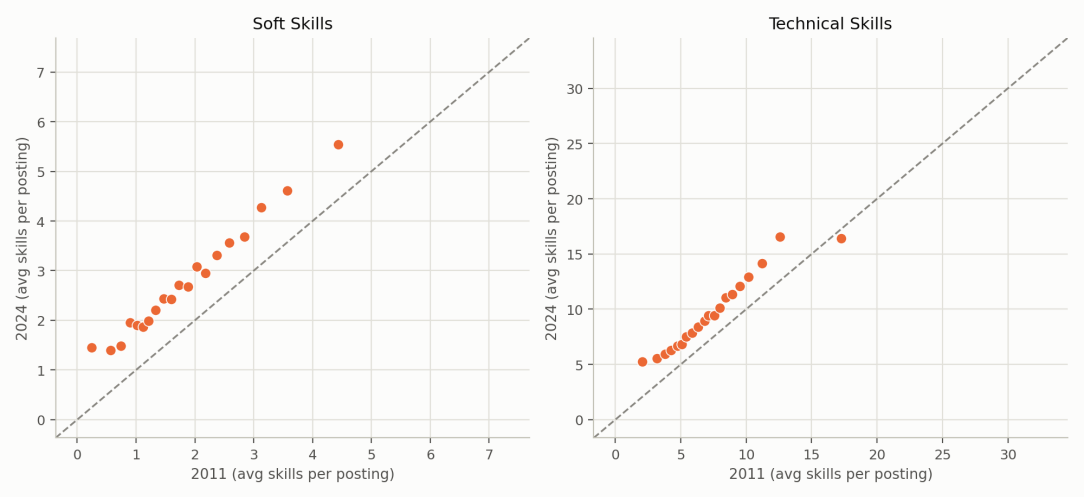


Finally, because all occupations experience a similar increase in skill demand (as observed in Figure A8), the relative skill requirements across these occupations also remains the same. If, for example, less soft-skill intensive occupations experienced a greater increase in soft skill

demand, we would have seen a flatter line formed by the points in the binned scatterplot in Figure A6 relative to the 45 degree line.

To summarize, the O*NET and Lightcast data combined paint an interesting picture about the nature of the rise in skill demand. It suggests that the core tasks of an occupation remain unchanged over time (as captured by O*NET data). However, the rise in soft skill demand implies that all occupations now require an additional component, which requires workers to communicate and coordinate effectively with the people in their workplace.

Figure A8: Lightcast Skill Requirements (2011-2024)



We now show that O*NET and Lightcast capture similar information for occupations in Table A5. We first classify O*NET occupations based on whether they had above median (High) or below median (Low) skill requirements in 2011. Our results are robust to using 2024 data as well. Then, we calculate the average number of skills required per job posting within these two groups using Lightcast data. Table A5 shows that for different skill dimensions (soft and technical) and skill measurement definitions (importance and level), the High group requires more skills per job posting than the Low group, while the last column shows that the rank correlation between the O*NET and Lightcast skill measures is positive as well. These comparisons assess cross-sectional agreement; they do not validate changes in required proficiency over time.

Overall, even though Lightcast data cannot measure the importance of different skills within an occupation like O*NET does, it still is able to differentiate between low and high-skill intensive occupations. Moreover, unlike O*NET, our Lightcast data is able to capture the rise in the demand for skills over time. Therefore, we choose to work with Lightcast data in our main analysis.

Cross-sectional comparison. As noted above, O*NET reports both the importance of a skill and the level of proficiency required.[23] We use its definition of soft skills for the skill classification and its occupational ratings to assess our measures of skill demand, as described above.

[23] See the O*NET overview and the scale definitions.

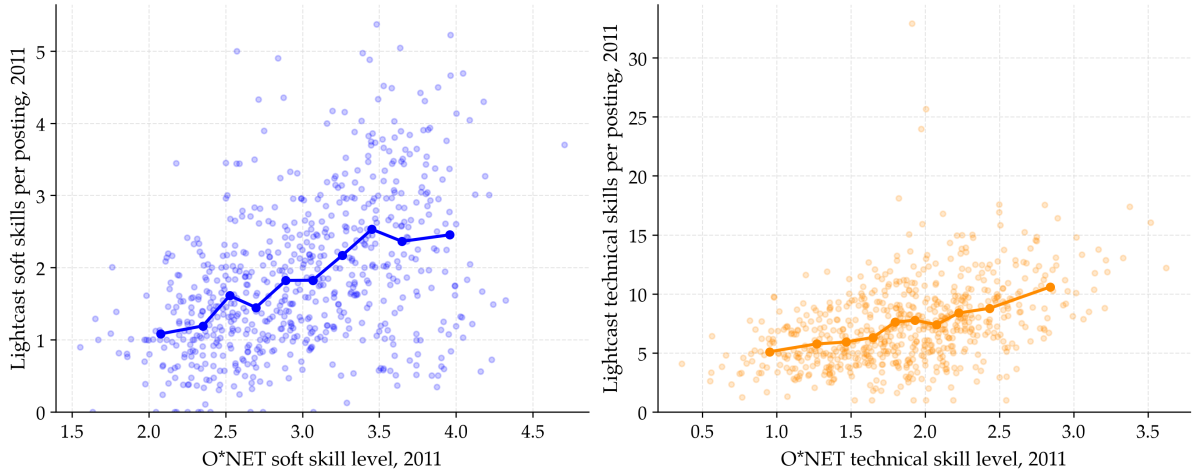
Table A5: O*NET vs. Lightcast: Construct Validity (2011)

O*NET Classification	Mean, High	Mean, Low	Difference	Rank Corr. (ρ)
Soft Importance	2.30	1.43	+0.87	0.43
Soft Level	2.28	1.45	+0.83	0.45
Technical Importance	8.34	6.31	+2.03	0.39
Technical Level	8.58	6.09	+2.49	0.47

High/Low = occupations above/below the 2011 O*NET median (Importance or Level, soft or technical, as indicated). Mean columns give the group's average number of Lightcast skills per posting in 2011. Rank Corr. is the Spearman correlation between the continuous O*NET score and the continuous Lightcast level, $N = 669$. Occupations receive equal weight. We do not report uncertainty for the High-vs-Low comparisons.

Figure A9 compares the Lightcast measures with O*NET skill-level ratings in 2011 for 669 matched occupations. Occupations with higher O*NET ratings generally have higher advertised skill counts. Table A5 reports comparisons using both importance and level ratings. These comparisons assess cross-occupation differences; they do not establish that changes in counts measure changes in required proficiency over time.

Figure A9: Advertised Skill Requirements and O*NET Occupational Ratings



Notes: Each light point is an occupation in 2011. The horizontal axis is its O*NET skill-level rating; the vertical axis is the average number of skills per Lightcast posting. The connected points show means within ten quantile groups of the O*NET rating, with equal weight for each occupation. Soft and technical requirements are shown separately. The sample is the intersection of 2011 Lightcast occupations and the harmonized O*NET classification panel. No causal interpretation is attached to the relationship, and the two sources measure different aspects of requirements.

A4 Additional Empirical Analysis

A4.1 Bachelor's Fields of Young College Graduates

Figure A10 shows how the bachelor's fields of young degree holders have changed since 2011. The shares with STEM and health-related degrees have risen, while the humanities, business, and social-science shares have fallen. These changes coincide with the erosion of young graduates' unemployment advantage documented in Section II; they do not directly measure workers' skills or establish the cause of that erosion.

We use annual national person microdata from the American Community Survey (ACS) for 2011–2024, excluding Puerto Rico. The sample consists of civilian, noninstitutional respondents aged 18–25 with a bachelor's degree or higher. Both sexes, enrolled graduates, unemployed graduates, and those outside the labor force are included. These are repeated cross-sections of young degree holders, not a panel of students or a count of degrees awarded in each year. The age restriction does not identify the year of graduation.

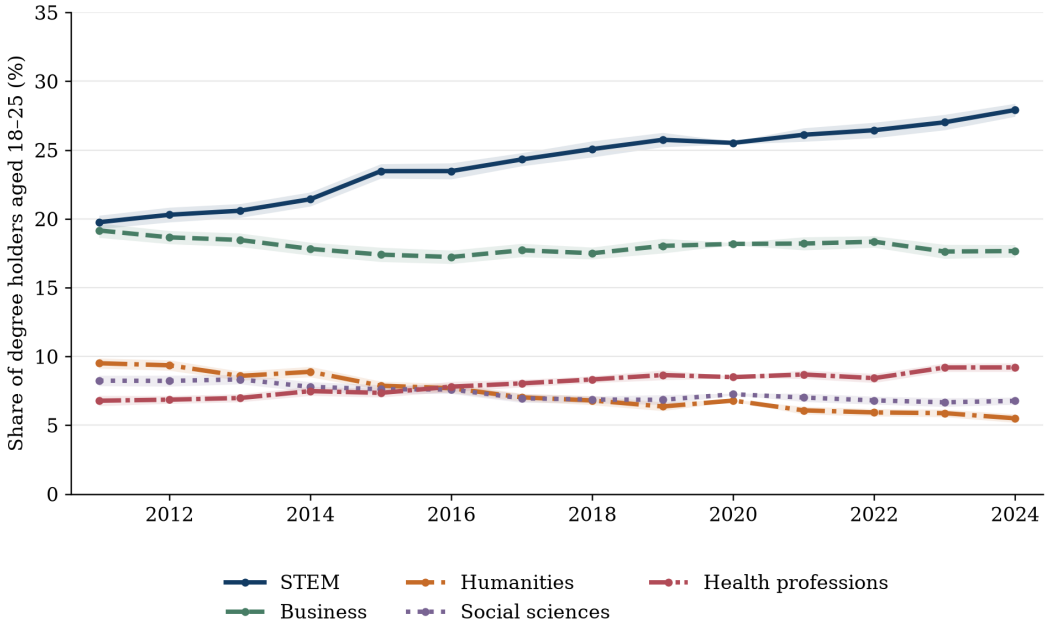
Each person counts once, using the first-listed bachelor's field (FOD1P), including for respondents with two bachelor's fields or an advanced degree. First-listed does not necessarily mean the primary field. Our core-STEM definition covers computing, engineering and engineering technologies, biology, mathematics/statistics, physical sciences and science technologies (Census prefixes 21, 24, 25, 36, 37, 50, and 51), plus mathematics/computer science, cognitive science/biopsychology, and data science/analytics (4005, 4006, and 4009). Health professions (61) is displayed separately and is not part of this STEM definition. Humanities combines area studies, languages, English, liberal arts/humanities, philosophy/religion, theology, and history (15, 26, 33, 34, 48, 49, and 64); it excludes arts. Business uses prefix 62. Social sciences uses prefix 55, including economics and sociology; psychology (52) remains outside the displayed groups. These groups are based on the Census field codes. They are not measures of workers' technical or soft skills.

For each year, a group's share is its weighted number of degree holders divided by the weighted number of all selected degree holders. Fields not displayed remain in the denominator. We use annual ACS person weights and all 80 person replicate weights; sampling variance is 4/80 times the sum of squared deviations of the replicate shares from the full-sample share. Figure bands are normal 95% confidence intervals. We include the 2020 one-year experimental release using its experimental person and replicate weights. No years are interpolated and the series are not smoothed. Detailed field codes can change over time; in particular, data science has a separate code only in later releases.

Table A6 reports the endpoint shares. STEM rose by 8 percentage points and health professions by 2 points. Humanities fell by 4 points, while business and social sciences each fell by about 2 points. The five displayed groups account for 63% of young degree holders in 2011 and 67% in 2024; remaining fields include education, psychology, arts, communications, and other applied fields. The graph therefore documents a shift toward STEM and health-related bachelor's fields among young degree holders, rather than an increase in their enrollment in these subjects.

Figure A11 plots young graduates' unemployment rates minus the overall rate among civilians aged 16–55, across all education levels, for the same five fields and arts (Census prefix 60). Zero denotes the overall rate. Over 2011–2024, unemployment in most of the fields shown, including STEM, increased relative to the overall rate. STEM graduates moved from below the overall rate to above it, while health professions remained below it. A relative increase need not imply an absolute increase in unemployment: a field's rate can fall while the overall rate falls faster. These comparisons describe outcomes within broad fields and do not identify the cause of the changes.

Figure A10: Bachelor's Fields of Young College Graduates



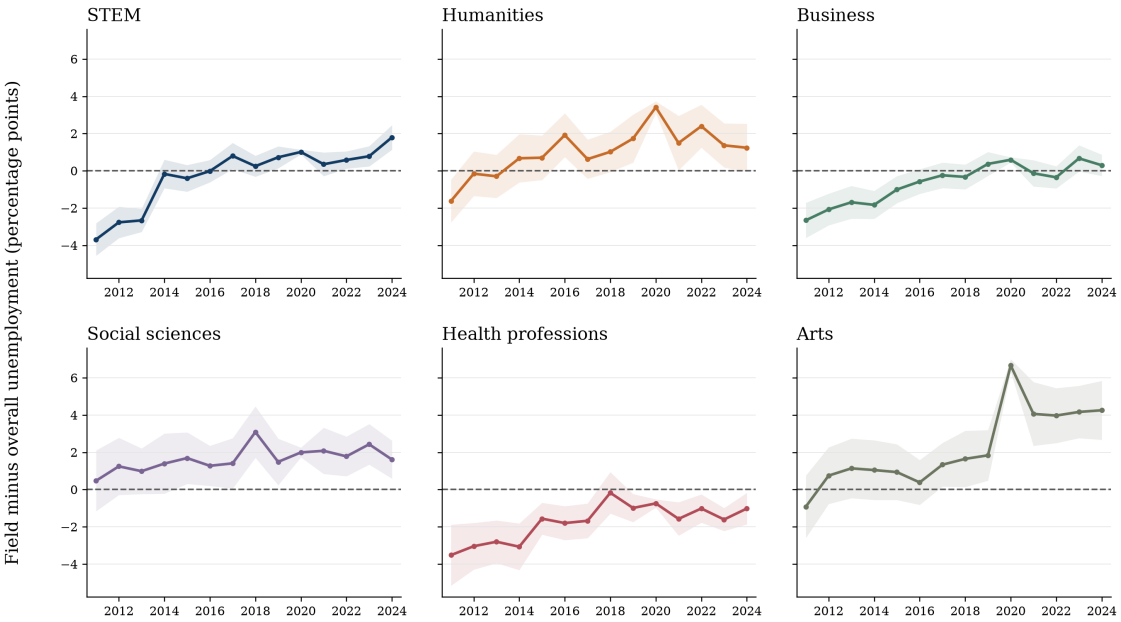
Notes: ACS one-year national person PUMS, 2011–2024; civilian, noninstitutional bachelor's-or-higher degree holders aged 18–25. Shares use the first-listed bachelor's field and divide by all selected degree holders. Shading shows 95% confidence intervals using the 80 person replicate weights. 2020 uses Census experimental weights. Field definitions and sample construction are described above.

Table A6: Changes in Bachelor's-Field Shares among Young Degree Holders

Field	2011 (%)	2024 (%)	Change (pp)
STEM	19.75	27.91	8.15
Humanities	9.50	5.50	−4.00
Business	19.15	17.66	−1.50
Social sciences	8.24	6.77	−1.47
Health professions	6.78	9.20	2.42
Other fields	36.57	32.97	−3.60

Notes: Sample, weights, and field definitions match Figure A10. Changes use unrounded shares; "Other fields" is the remainder and is not displayed in the figure.

Figure A11: Unemployment among Young College Graduates by Bachelor's Field

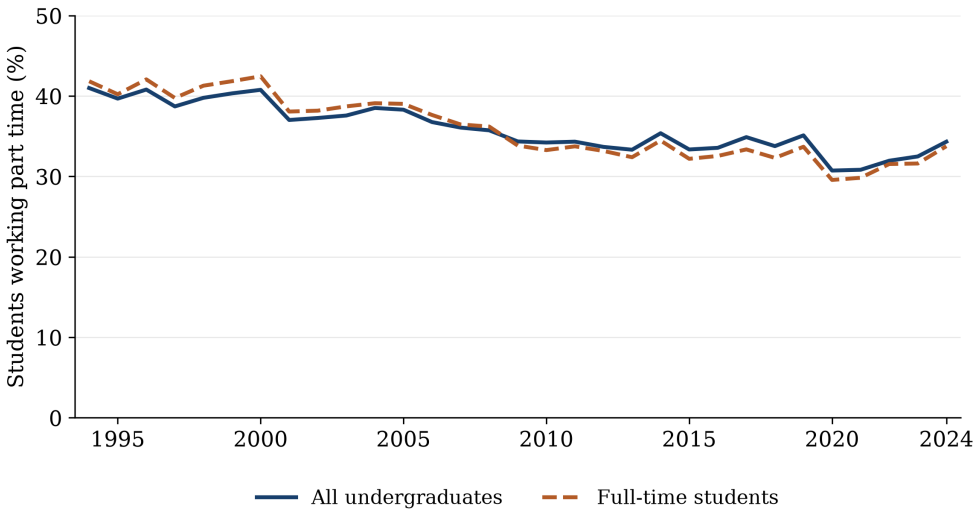


Notes: ACS one-year national person PUMS, 2011–2024. Lines show each field's unemployment rate minus the overall rate, in percentage points. The dashed zero line denotes equality with the overall rate. Field samples are civilian, noninstitutional bachelor's-or-higher degree holders aged 18–25, classified by their first-listed bachelor's field. Definitions match Figure A10, with arts (Census prefix 60) added. The overall benchmark includes civilian, noninstitutional respondents aged 16–55 across all education levels. Rates divide weighted unemployed respondents by weighted employed plus unemployed respondents; those outside the labor force are excluded from this denominator. Employment includes self-employment. Shading shows 95% confidence intervals for the gaps using the 80 person replicate weights, retaining covariance with the overall benchmark. 2020 uses Census experimental weights. Annual series are not smoothed.

A4.2 Part-Time Work among College Students

Figure A12 shows the share of undergraduates aged 18–24 working part time. This share fell from 41% in 1994 to 34% in 2024. Most of the decline occurred before 2011; the shares in 2011 and 2024 are nearly identical. The series restricted to full-time students follows a similar pattern.

Figure A12: Part-Time Employment among College Students



Notes: October CPS School Enrollment Supplement, 1994–2024. The sample includes civilian undergraduates aged 18–24 in two- and four-year colleges; graduate students are excluded. Each line divides students who usually work fewer than 35 hours per week across all jobs by all students in the relevant enrollment group, including those who are not employed. The dashed line restricts the sample to students enrolled full time. Employment includes self-employment and temporary absences from work. Estimates use CPS final weights through 2005 and supplement weights from 2006 onward. Annual estimates are not smoothed. The October observations do not measure summer work.

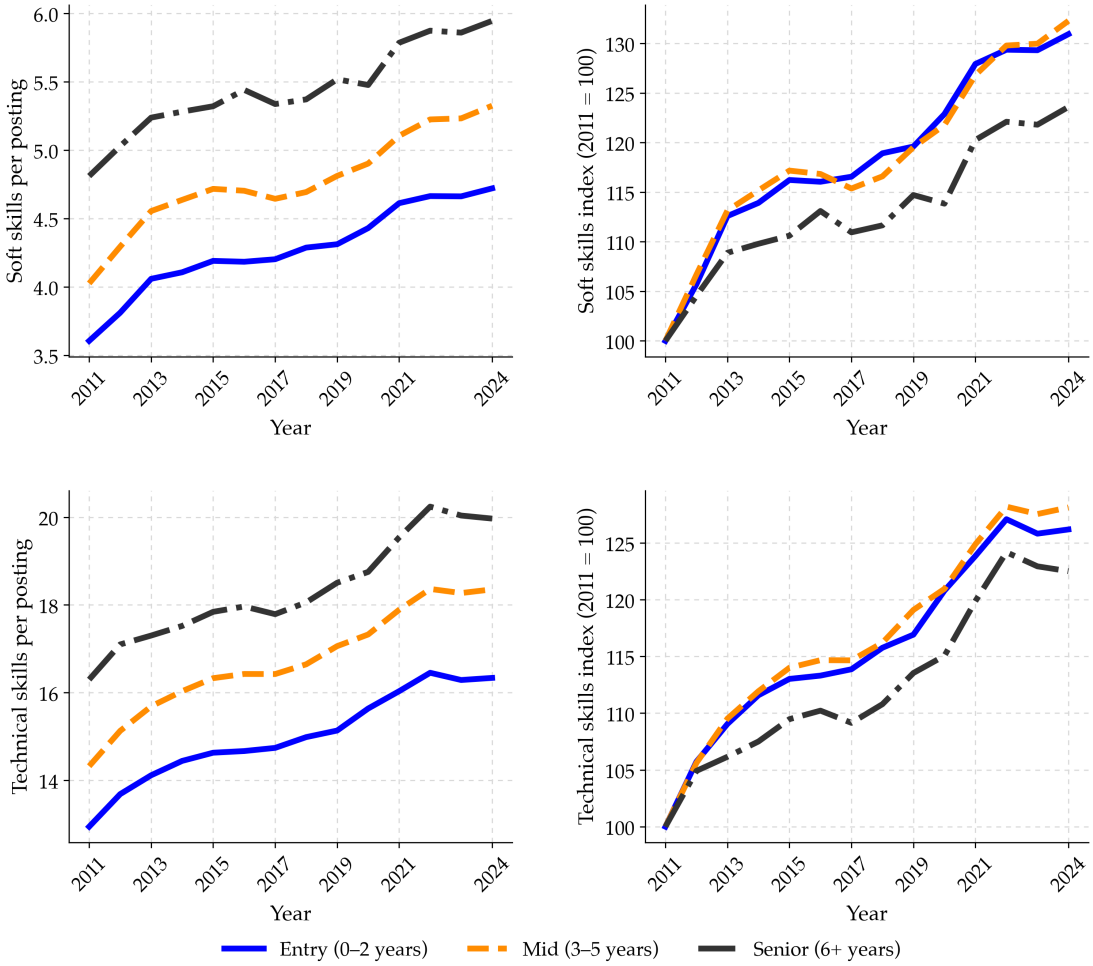
A4.3 Skill Requirements by Required Work Experience

Figure A13 compares the level and growth of skill requirements across entry, mid-level, and senior postings, holding occupational composition fixed. The groups refer to the minimum experience stated in an advertisement, rather than the age or experience of the worker ultimately hired.

A4.4 Unemployment Rate of Non-College Graduates

Figure A14 compares young noncollege workers' unemployment rate with the overall rate. Their gap narrows, whereas young college graduates lose their earlier unemployment advantage (Figure 2). Figure A15 plots the noncollege gap using the same sample and overall benchmark as the main-text figure.

Figure A13: Skill Requirements by Required Work Experience

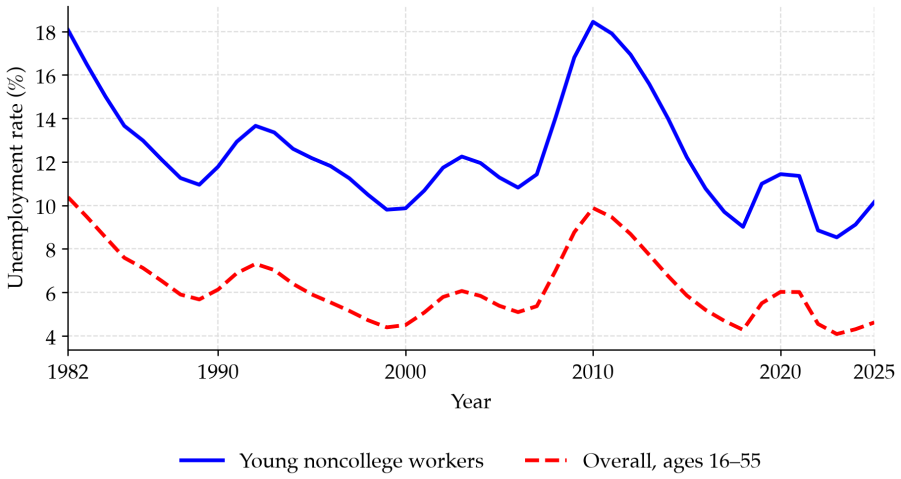


Notes: The sample consists of Lightcast postings requiring a bachelor's degree or higher and classified as full-time (more than 32 hours), 2011–2024. Entry, mid-level, and senior postings require 0–2, 3–5, and at least 6 years of experience, respectively. Postings with missing experience are excluded. The left panels show average skill counts per posting; the right panels index each series to 100 in 2011, rather than reporting annual growth rates. Soft skills appear in the top row and technical skills in the bottom row. Skill and posting counts are summed within occupation, experience group, and year before calculating ratios. All series use the same 473 occupations, observed in every group and year, and fixed occupation weights based on pooled posting shares across groups and years. The common sample contains 55,651,268 postings. The series are not adjusted for posting length and do not measure workers' skill acquisition.

A4.5 Returns To Soft And Technical Skills For Young Workers

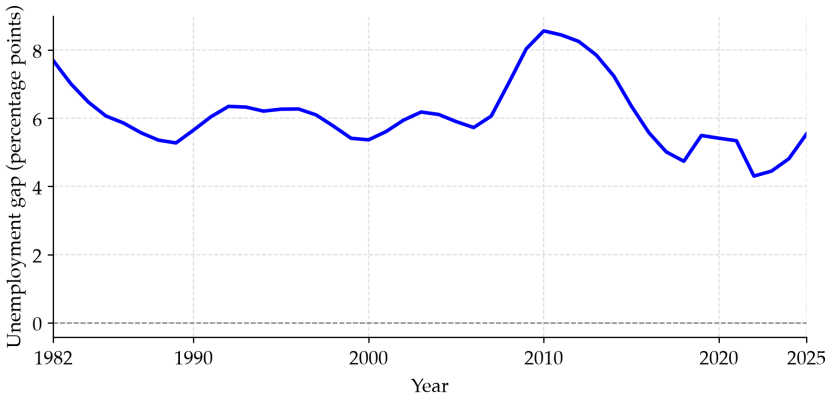
The soft-skill interaction is negative in each specification in Table A7. For college workers it is about −0.03 log points per additional soft skill across specifications; the age-control estimate has $p = 0.064$. The college technical-skill interaction is essentially zero when occupation fixed effects are included. The two skill dimensions therefore do not support equally stable conclusions across specifications. These estimates describe wage associations among employed workers and do not identify how workers acquire skills or the effect of skill requirements on unemployment.

Figure A14: Young Non-College Workers vs. Overall Unemployment Rate



Notes: Annual CPS unemployment rates, 1982–2025, using the sample and smoothing in Figure 2. Young workers are aged 16–25; the overall benchmark covers ages 16–55. Noncollege workers do not hold a bachelor's degree. Rates use final CPS survey weights and include the self-employed in the labor force.

Figure A15: Gap Between The Young Non-College Workers vs. Overall Unemployment Rate



Notes: Young noncollege workers' unemployment rate minus the overall rate, using the CPS sample and smoothing in Figure 2. Young workers are aged 16–25 and the overall sample covers ages 16–55. Both rates include the self-employed in the labor force and use final CPS weights.

A4.6 Entry-Level Requirements and Within-Occupation Wage Analysis

We next measure requirements using only full-time postings requiring at least a bachelor's degree and 0–2 years of experience. We sum skill mentions and postings within harmonized occupation–year cells before calculating average requirements. This preserves the taxonomy and the occupation crosswalk used above. The wage sample, demographic controls, and CPS weights are unchanged except that workers in cells without entry-level requirements are excluded.

Table A8 compares the baseline with the entry-level specification. The interaction between Young and soft requirements is −0.0240 (SE 0.0114, $p = 0.036$): one additional soft skill per entry-level posting is associated with a 0.024-log-point reduction in young graduates' wages relative to older graduates. The baseline all-posting coefficient is −0.0300. The entry-level

Figure A16: Wage Gap Regression Coefficients Over Time

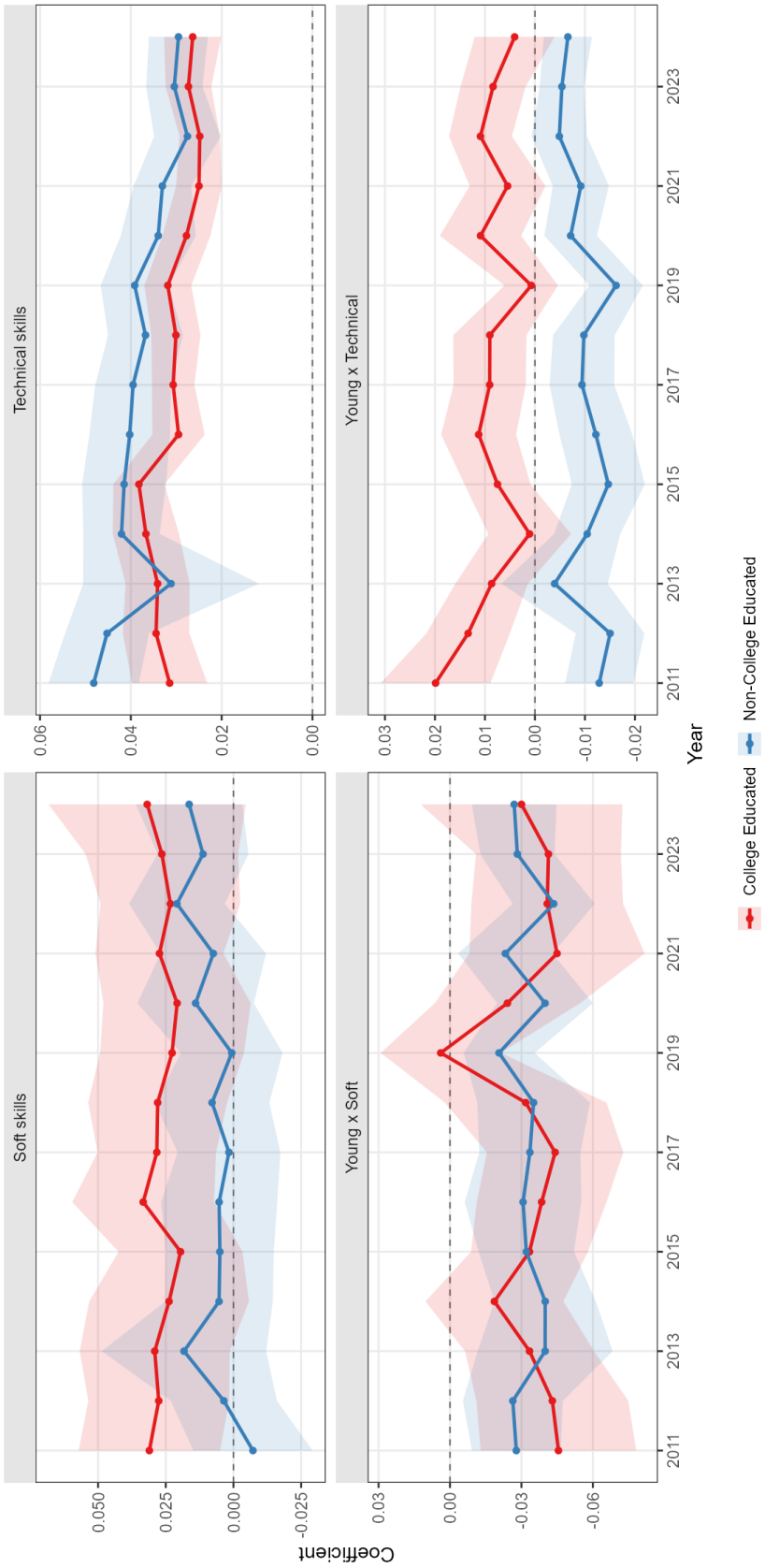


Table A7: Young–Older Wage Differences and Skill Requirements: Robustness

	College		Noncollege	
	Young × Soft	Young × Tech.	Young × Soft	Young × Tech.
Baseline	-3.00**	0.93***	-2.52***	-0.80***
	(1.42)	(0.33)	(0.95)	(0.27)
Fixed occupation requirements	-3.29**	0.78**	-3.45***	-0.92***
	(1.58)	(0.32)	(1.10)	(0.30)
Occupation fixed effects	-3.25***	0.00	-1.73***	-1.40***
	(0.69)	(0.18)	(0.66)	(0.29)
Age and age squared	-2.61*	0.85***	-2.22***	-1.10***
	(1.41)	(0.33)	(0.81)	(0.26)
Observations	572,091		931,070	

Notes: The outcome is log real hourly earnings. Coefficients and standard errors are multiplied by 100; a coefficient is 100 times the change in the young–older log wage gap associated with one additional advertised skill per posting, so that −3.00 denotes a change of −0.03 log points. The sample covers 2011–2024 and ages 16–54; young denotes age 25 or younger and college denotes a bachelor's degree or higher. These specifications differ from the separate ages 21–29 labor market flow analysis. All regressions include both skill measures, their interactions with young, sex, race, education, and calendar-month fixed effects. They use composition-adjusted final CPS weights and occupation-clustered standard errors. Each alternative changes only the stated feature of the baseline. Fixed occupation requirements pool mentions and postings over 2011–2024. Occupation fixed effects absorb occupation mean wages; they do not absorb occupation-specific age wage gaps. The age specification adds age and age squared. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

technical interaction is positive but imprecise, at 0.0056 (SE 0.0035, $p = 0.106$). Entry-level measures retain 569,719 of the baseline 572,091 respondents, 421 of 437 occupations, and 5,305 of 5,691 occupation–year cells. Using annual all-posting requirements on the baseline sample gives a soft interaction of −0.0293 (SE 0.0146), so annual aggregation alone does not account for the difference from the baseline.

Persistent differences across occupations can still explain the wage gap in these regressions. To examine changes within occupations, we estimate

$$\log w_{iot} = \alpha_{ot} + \gamma_o \text{Young}_i + \lambda_t \text{Young}_i \tag{36}$$

$$+ \beta_s(\text{Young}_i \times \text{Soft}_{ot}) + \beta_T(\text{Young}_i \times \text{Tech}_{ot}) + X_i'\theta + \varepsilon_{iot}, \tag{37}$$

where t now denotes calendar years. Occupation–year fixed effects absorb common wage changes within each occupation and the main effects of requirements. Young interacted with occupation absorbs persistent occupation-specific wage gaps; Young interacted with year absorbs national changes in the gap. The skill interactions are therefore identified by occupation-specific changes in requirements relative to common year changes, rather than by permanent differences in requirements across occupations. Standard errors are clustered by harmonized occupation.

The within-occupation estimates do not show the predicted relative deterioration. The soft interaction is 0.0112 (SE 0.0142) with all postings and 0.0094 (SE 0.0073) with entry-level postings; neither differs statistically from zero. The 95% confidence interval is $[-0.0167, 0.0391]$ for all postings and $[-0.0050, 0.0239]$ for entry-level postings. The technical interactions are also insignificant. Restricting the sample to 2011–2019, requiring at least 30, 100, or 500 postings per occupation–year, or holding observable advertisement composition fixed does not

Table A8: Entry-Level Skill Requirements and the Young–Older Wage Gap

	All postings Baseline (1)	Entry-level Baseline (2)	All postings Within occupation (3)	Entry-level Within occupation (4)
Young × Soft	-0.0300**	-0.0240**	0.0112	0.0094
	(0.0142)	(0.0114)	(0.0142)	(0.0073)
Young × Technical	0.0093***	0.0056	0.0073	-0.0002
	(0.0032)	(0.0035)	(0.0054)	(0.0028)
Young	-0.3724***	-0.3617***	Absorbed	Absorbed
	(0.0391)	(0.0654)		
Soft	0.0266*	0.0161	Absorbed	Absorbed
	(0.0152)	(0.0126)		
Technical	0.0296***	0.0305***	Absorbed	Absorbed
	(0.0031)	(0.0026)		
Skill-measure frequency	Monthly	Annual	Annual	Annual
Calendar-month FE	Yes	Yes	No	No
Occupation × year FE	No	No	Yes	Yes
Young × occupation and year FE	No	No	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Observations	572,091	569,719	572,091	569,719
Young respondents	47,884	47,653	47,884	47,653
Occupations	437	421	437	421
Occupation–year cells	5,691	5,305	5,691	5,305

Notes: The outcome is log real hourly earnings. Young denotes age 25 or younger; older workers are aged 26–54. All columns use the baseline CPS MORG college-graduate sample (ages 16–54), composition-adjusted final weights, and indicators for sex, race, and degree. Column (1) reproduces Table 3's monthly point estimates; its recalculated occupation-clustered SEs are independently confirmed in Stata and differ slightly from those reported in that table (soft: 0.014244 versus 0.014166; technical: 0.003242 versus 0.003333). Entry measures use full-time postings requiring at least a bachelor's degree and explicitly 0–2 years of experience. Mentions and postings are summed across linked SOC codes before taking ratios. Missing requirements are excluded, never set to zero. Columns (3)–(4) absorb occupation–year levels and occupation-specific and year-specific young–older gaps. Skill main effects and Young are absorbed. SEs in parentheses cluster by harmonized occupation, using CR1 with the full fixed-effect rank and $t(G-1)$ inference. Coefficients are log-wage changes per additional skill per posting. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

produce a significant negative soft interaction (Tables A9–A10). Table A10 also documents sensitivity to posting length and to a small number of postings omitted from the entry-level denominators.

The entry-level baseline thus supports the cross-occupation wage pattern, but the changes within occupations do not establish a stronger link between rising soft requirements and worsening relative wages. These associations do not identify a causal effect of requirements or establish that soft skills are accumulated on the job.

Table A9 reports the within-occupation specification for the full and pre-COVID periods and alternative posting-count thresholds. Table A10 separates changes in measurement frequency, respondent coverage, and observable advertisement composition. The common-strata and fixed-mix entry specifications use the same detailed support, so their comparison isolates reweighting from the support restriction. Neither yields a significant negative soft-skill interaction. These restrictions cannot account for unobserved changes in employers or job tasks within a cell.

Re-estimating the baseline reproduces the wage coefficients in Table 3. The recalculated clustered standard errors differ slightly from those reported there; independent Stata estimation agrees with the recalculation. Independent explicit-dummy regressions also confirm the within-occupation coefficients and covariance calculation. Simulation stress tests using the observed occupation–year support recover the coefficients but show some undercoverage of conventional clustered confidence intervals. These simulations compress within-cell demographic variation and are an inference diagnostic, not a validation of causal identification. We report conventional occupation-clustered inference and treat the within-occupation results as inconclusive.

Table A9: Within-Occupation Wage Results: Period and Posting-Count Restrictions

Years	Minimum ads	Young × Soft	Young × Tech.	*N*	Occ.	Cells
A. All postings						
2011–2024	1	0.0112 (0.0142)	0.0073 (0.0054)	572,091	437	5,691
2011–2024	30	0.0105 (0.0143)	0.0077 (0.0055)	571,731	432	5,601
2011–2024	100	0.0065 (0.0142)	0.0085 (0.0056)	570,699	423	5,494
2011–2024	500	0.0027 (0.0142)	0.0073 (0.0057)	561,325	407	5,239
2011–2019	1	0.0057 (0.0170)	0.0050 (0.0079)	383,061	435	3,686
2011–2019	30	0.0054 (0.0172)	0.0056 (0.0080)	382,802	430	3,615
2011–2019	100	0.0036 (0.0176)	0.0064 (0.0082)	381,818	420	3,522
2011–2019	500	-0.0011 (0.0181)	0.0019 (0.0083)	372,959	401	3,325
B. Entry-level postings						
2011–2024	1	0.0094 (0.0073)	-0.0002 (0.0028)	569,719	421	5,305
2011–2024	30	0.0106 (0.0082)	-0.0016 (0.0048)	549,232	343	4,198
2011–2024	100	0.0082 (0.0099)	-0.0024 (0.0056)	525,069	289	3,373
2011–2024	500	0.0004 (0.0109)	0.0070 (0.0063)	469,724	203	2,230
2011–2019	1	0.0078 (0.0079)	-0.0030 (0.0031)	381,309	416	3,406
2011–2019	30	0.0123 (0.0090)	-0.0065 (0.0059)	365,325	327	2,624
2011–2019	100	0.0088 (0.0103)	-0.0102* (0.0061)	346,705	266	2,068
2011–2019	500	0.0076 (0.0120)	-0.0041 (0.0077)	307,946	182	1,344

Notes: Every column uses the within-occupation specification in Table A8, with both skill interactions, occupation–year fixed effects, Young interacted with occupation and year, and the baseline demographic controls and CPS weights. Minimum ads is the number of postings used to construct that occupation–year's skill measure; a threshold of one is the unrestricted positive-count sample. Thresholds and missing exposures remove respondents; they do not replace exposures with zero. Occupations are inference clusters. Coefficients and occupation-clustered SEs (in parentheses) are in log-wage units per additional advertised skill. CR1 counts the full fixed-effect rank; inference uses $t(G-1)$. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

Table A10: Wage Results: Aggregation, Coverage, and Advertisement Composition

Specification	Young × Soft	Young × Tech.	*N*	Occ.
Annual all-posting baseline	-0.0293**	0.0089***	572,091	437
	(0.0146)	(0.0033)		
Annual baseline, entry-covered sample	-0.0292**	0.0090***	569,719	421
	(0.0147)	(0.0033)		
College/full-time, known experience: baseline	-0.0329***	0.0040	570,520	428
	(0.0121)	(0.0035)		
All-posting within, entry-covered sample	0.0051	0.0090	569,719	421
	(0.0142)	(0.0055)		
College/full-time, known experience: within	0.0009	-0.0026	570,520	428
	(0.0080)	(0.0039)		
All-posting within, fixed SOC mix	0.0138	0.0066	572,076	435
	(0.0140)	(0.0053)		
Entry within, common detailed strata	0.0072	0.0007	564,612	338
	(0.0077)	(0.0028)		
Entry within, fixed detailed-stratum mix	0.0071	0.0021	564,612	338
	(0.0086)	(0.0025)		
All-posting within, word-count interaction	0.0074	0.0043	572,091	437
	(0.0141)	(0.0057)		
Entry within, expanded posting denominator	0.0095	-0.0002	569,719	421
	(0.0074)	(0.0028)		

Notes: Baseline and within specifications follow Table A8. The first two rows separate annual aggregation and exposure coverage from the entry-level restriction. Known experience uses all experience levels within the college/full-time stated-experience sample. Fixed mixes use pooled posting shares, retaining detailed strata observed in every year, 2011–2024. For all postings, strata are SOC codes within a harmonized CPS occupation; for entry postings they are SOC × degree requirement × exact required experience (0, 1, or 2 years). The common-strata row uses actual annual shares on that same support. The word-count row adds Young interacted with occupation–year average words per posting; the main effect is absorbed. Word counts are unavailable separately for entry postings. The denominator check adds 1,440 postings in known numerator cells omitted when matching on exact required experience (0.010% of entry postings); four occupation–years lacking numerator records remain missing. The taxonomy is unchanged. These checks cannot fix unobserved employer, job-task, or reporting composition. SEs cluster by occupation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A5 Labor Market Flows of Young College Graduates

A deterioration in young graduates' relative labor market outcomes can arise through slower job finding, more frequent separations, or changes in labor force participation. Using matched monthly CPS observations, we examine these margins over 2011–2024. We compare college graduates aged 21–29 with graduates aged 30–55 and workers aged 21–29 without a bachelor's degree. Age and education are measured before the transition. Employment includes both wage work and self-employment. Each transition probability conditions on the worker's initial labor market state.[24]

Young college graduates lose their initial advantage in finding employment from unemployment. Figure A17 shows the annual series, and Table A11 reports period averages. In 2011–2015, 28% of unemployed young graduates were employed the following month, compared with 21% of older graduates and 20% of young workers without a college degree. By 2022–2024, all three rates were about 28%. Young graduates' job-finding rate was nearly unchanged across these periods, while both comparison groups improved. Their advantage over older graduates had already narrowed from 6 to 2 percentage points by 2016–2019, before the pandemic.

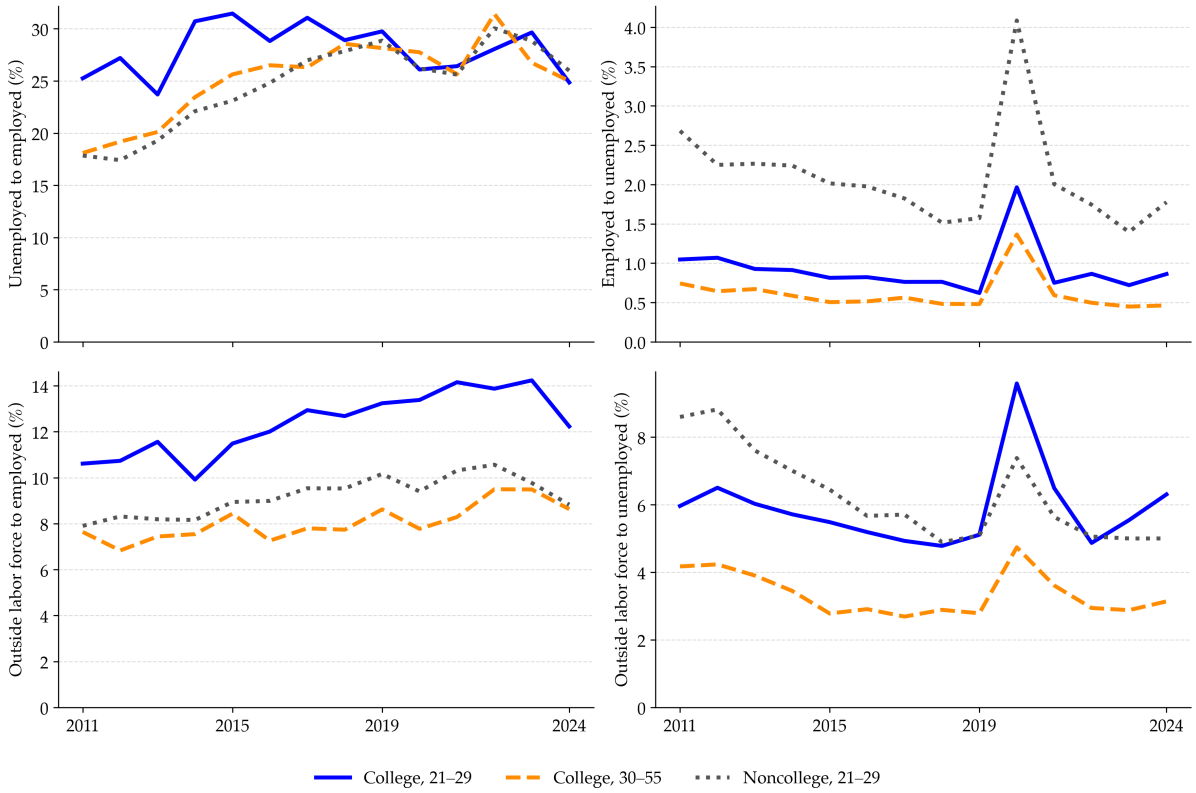
Separations into unemployment show little change in young graduates' position relative to older graduates. The monthly employment-to-unemployment probability declined for both young and older graduates, remaining less than 1% for both groups in each period. The gap between them remained close to one-third of a percentage point. Young graduates also continued to have lower separation rates than young workers without a degree, although this difference narrowed. These transitions include both voluntary and involuntary separations.

The participation flows place a further limit on the interpretation. Among young graduates outside the labor force, the probability of moving directly into employment rose from 11% to 13%. These moves include both first entry and reentry, so they do not measure school-to-work transitions alone. The erosion of young graduates' relative job finding is therefore specific to those starting from unemployment; it does not extend to all routes into employment.

The decline in young graduates' job-finding advantage motivates examining why they have not shared in the improvement experienced by other unemployed workers. Changes in the composition of unemployment and differences in responses to the business cycle can also contribute to this pattern. The comparisons are descriptive, without confidence intervals, and do not identify the contribution of changing skill requirements to relative unemployment.

[24] We link respondents using the validated CPS person identifier, requiring consecutive calendar months and the appropriate progression through rotation groups. We exclude ASEC records and observations with missing labor market states or education, invalid demographic matches, or nonpositive longitudinal weights. Monthly rates use the origin month's forward longitudinal weight, `LNKFW1MWT`. Annual and period estimates give each month equal weight. The age range differs from the main text's definition of young workers as those aged 25 or younger.

Figure A17: Labor Market Transitions by Age and Education



Notes: Annual averages of monthly transition probabilities in the matched CPS, 2011–2024. Each rate is the longitudinally weighted share of respondents in the initial state who enter the destination state the following month. College denotes a bachelor's degree or higher; employment includes self-employment.

Table A11: Monthly Labor Market Transition Probabilities

	2011–2015	2016–2019	2020–2021	2022–2024
Unemployed → employed				
College, 21–29	27.67	29.63	26.25	27.52
College, 30–55	21.29	27.38	26.70	27.72
Noncollege, 21–29	19.95	27.13	25.90	28.29
Employed → unemployed				
College, 21–29	0.95	0.74	1.36	0.82
College, 30–55	0.63	0.51	0.98	0.47
Noncollege, 21–29	2.29	1.72	3.04	1.64
Outside labor force → employed				
College, 21–29	10.86	12.72	13.77	13.45
College, 30–55	7.58	7.85	8.03	9.20
Noncollege, 21–29	8.30	9.55	9.86	9.72
Outside labor force → unemployed				
College, 21–29	5.94	5.00	8.04	5.57
College, 30–55	3.71	2.82	4.17	2.99
Noncollege, 21–29	7.69	5.34	6.51	5.02
Employed → outside labor force				
College, 21–29	1.94	1.93	2.01	1.97
College, 30–55	1.09	1.13	1.24	1.18
Noncollege, 21–29	3.46	3.63	4.24	3.74
Unemployed → outside labor force				
College, 21–29	16.76	18.95	18.65	18.07
College, 30–55	15.07	17.43	15.76	17.43
Noncollege, 21–29	23.85	25.71	24.70	25.04

Notes: Entries are percentages, averaged across months within each period. The matched CPS sample and weighting follow Figure A17. Each probability conditions on the initial state indicated by the row heading. The table reports all six transitions among employment, unemployment, and nonparticipation; pandemic years are shown separately.